

An applicable and efficient retrograde monosynaptic circuit mapping tool for larval zebrafish

Tian-Lun Chen, Qiu-Sui Deng, Kun-Zhang Lin, Xiu-Dan Zheng, Xin Wang, Yong-Wei Zhong, Xin-Yu Ning, Ying Li, Fu-Qiang Xu, Jiu-Lin Du , Xu-Fei Du 

Institute of Neuroscience, State Key Laboratory of Neuroscience, Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, China • University of Chinese Academy of Sciences, Beijing, China • The Brain Cognition and Brain Disease Institute (BCBDI), Shenzhen Key Laboratory of Viral Vectors for Biomedicine, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China • School of Life Science and Technology, ShanghaiTech University, Shanghai, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This **important** study offers substantial technical advancements for neural circuit tracing in larval zebrafish, a model for systems and developmental neurobiology. The enhanced rabies virus-based retrograde transneuronal tracing improves efficiency and provides a method for combined structural and functional brain mapping. The supporting evidence is **solid**, and there is strong confidence in the technique's utility for neurobiologists working with zebrafish.

<https://doi.org/10.7554/eLife.100880.2.sa3>

Abstract

The larval zebrafish is a vertebrate model for in vivo monitoring and manipulation of whole-brain neuronal activity. Tracing its neural circuits still remains challenging. Here we report an applicable methodology tailored for larval zebrafish to achieve efficient retrograde trans-monosynaptic tracing from genetically defined neurons via EnvA-pseudotyped glycoprotein-deleted rabies viruses. By combinatorially optimizing multiple factors involved, we identified the CVS strain trans-complemented with advanced expression of N2cG at 36°C as the optimal combination. It yielded a tracing efficiency of up to 20 inputs per starter cell. Its low cytotoxicity enabled the viable labeling and calcium imaging of infected neurons 10 days post-infection, spanning larval ages commonly used for functional examination. Cre-dependent labeling was further developed to enable input cell-type-specific tracing and circuit reconstruction. We mapped cerebellar circuits and uncovered the ipsilateral preference and subtype specificity of granule cell-to-Purkinje cell connections. Our method offers an efficient way for tracing neural circuits in larval zebrafish.

Introduction

Given the small transparent brain and accessible genetic manipulations, the larval zebrafish has been emerging as a promising animal model for systems neuroscience. Large-scale molecular, light- and electron-microscopical (LM and EM) morphological, and physiological datasets have been gathered over the past decade to decipher the identity and connectivity of its whole-brain cells^{1–7}. Whole-brain neurons' activity can be monitored and manipulated in awake as well as behaving zebrafish larvae, providing an unprecedented research paradigm for dissecting neural mechanisms of brain functions^{8–14}. Further mechanistic insights into the roles of different cell types in brain-wide activity require effective tools for dissecting underlying neural circuits¹⁵.

Virus-based tracing tools have opened new avenues for disentangling neural circuits^{16,17}. Among them, the EnvA-pseudotyped glycoprotein (G)-deleted rabies virus (RV) (RVdG[EnvA]) has achieved the most success^{18–20}. Its success derives from being a native retrograde transsynaptic tracer engineered for cell-type-specific targeting through the recognition between EnvA and its receptor TVA²¹ and crossing only one synaptic step via trans-complementation of rabies G^{18,19}. This tool has revolutionized the structural analysis of neural circuits in rodents^{15,22} and has continually evolved to adapt to functional studies^{23–25}. However, its application in other species including zebrafish is retarded due to a lack of tailored methodologies.

The G-deleted RV (RVdG) infection of neurons in zebrafish was first demonstrated in 2009. Ten years later, RVdG[EnvA] was applied to trace input neurons in the adult zebrafish cerebellum, but the efficiency showed very low²⁶. Recently, the herpes simplex virus (HSV1) was used as a helper virus to deliver TVA and rabies G, leading to an increased efficiency of around one input per starter neuron in the brain of adult zebrafish²⁷. For larval zebrafish, efforts have been focused on the vesicular stomatitis virus (VSV), another rhabdovirus similar to RV but with anterograde spread^{28–30}. However, these reported VSV tools for larval zebrafish appeared relatively high cytotoxicity and have not been iterated to utilize the EnvA-TVA system to achieve initial infection specificity^{28–30}. Therefore, developing applicable efficient viral tracing methods is still an important mission in the field, and will boost the application of larval zebrafish in systems neuroscience research.

In the present study, we established applicable methodologies to implement RVdG[EnvA] for efficient retrograde trans-monosynaptic tracing from genetically defined neurons in larval zebrafish. We first demonstrated that the transient co-expression of the helper proteins TVA and rabies G in specific neurons through one-cell-stage microinjection of the GAL-UAS binary DNA plasmids could successfully help the RVdG[EnvA] microinjected later on to infect and spread from targeted starter cells. This simple two-round microinjection procedure ensures the high feasibility of our method. Then via practicing in the cerebellar circuit, we iteratively tested key factors that influence the tracing efficiency and cytotoxicity, which include the virus strain, G protein type, G expression level, and working temperature. We discovered an optimal combinatory condition: CVSdG trans-complemented with advanced expression of N2cG at 36°C. The optimality lies in the high tracing efficiency and low cytotoxicity, embodied by a 20-fold increase in efficiency compared to previous studies in zebrafish^{26,27} and a long enough time window (final fish age up to 2–3 weeks old) for conducting functional studies in larval zebrafish. Furthermore, by using Cre-expressing RVdG, we developed a transgenic reporter framework to enable input cell-type-specific tracing and circuit reconstruction based on single-neuron morphology. As an application demonstration, we applied the method to map the monosynaptic inputs to Purkinje cells (PCs) from granule cells (GCs) in the cerebellum and revealed the wiring relationship between morphological subtypes of GCs and PCs. These endeavors prove the applicability and efficiency of our method in dissecting interested circuits in larval zebrafish.

Results

Implementation of RVdG[EnvA]-based neural circuit tracing in larval zebrafish

To implement RVdG[EnvA]-based retrograde tracing in larval zebrafish, we first verified that larval zebrafish themselves do not express endogenous TVA. In wild type (WT) larvae without exogenous TVA expression, the injected SADdG-mCherry[EnvA], a type of RVdG[EnvA], did not infect any cells (**Figure 1—figure supplement 1A** [↗](#)). The observed mCherry-positive puncta co-localized with microglia (**Figure 1—figure supplement 1B** [↗](#)), suggesting the clearance of uninfected virus particles by microglia. These data demonstrate that similar to mammals³¹ [↗](#), zebrafish do not express endogenous TVA.

Then we specifically expressed TVA in neurons (**Figures 1A and B** [↗](#)), and found that injected SADdG-mCherry[EnvA] infected TVA-expressing cells (**Figure 1C and D** [↗](#), dashed yellow circles). The targeted expression of TVA and the fluorescent indicator enhanced green fluorescent protein (EGFP) in neurons was achieved through transient transgene of GAL4-UAS constructs in which the GAL4 activator is driven by *elavl3*, a pan-neuronal marker in zebrafish (**Figure 1A** [↗](#), and “*UGT*” in **Figure 1B** [↗](#)). As the G protein is necessary for viral spread, SADdG-mCherry[EnvA] did not spread beyond the initially infected neurons (i.e. starter cells) (**Figure 1C,E–G** [↗](#)).

We next examined whether viral particles can spread to other cells by complementing the G protein in starter neurons. As the efficiency of virus spread is reported to be positively correlated with the level of G expression in starter neurons^{32,33} [↗](#), we generated two helper plasmids, *UAS:EGFP-P2A-TVA-T2A-B19G* (*UGTB*) and *UAS:EGFP-P2A-B19G-T2A-TVA* (*UGBT*), to search for a plasmid design with higher G expression while ensuring the co-expression of TVA and G (**Figure 1B** [↗](#), see Methods). The position of the RV G-protein-coding gene SAD B19G was varied in the tricistronic 2A constructs for adjusting its expression level³⁴ [↗](#). As shown in **Figure 1D** [↗](#), besides starter cells (EGFP⁺ and mCherry⁺, dashed yellow circles) and uninfected TVA-expressing cells (EGFP⁺ only), we observed mCherry⁺ only cells (arrows) of which the number increased over time, indicating the virus spread beyond the starter cells by trans-complementation with G. Notably, the traced cells included both neurons and radial glia (hereafter referred to as glia) identified by their distinct morphology (**Figure 1D** [↗](#), red and gray arrows, respectively). Notably, the trans-infection of glia was also observed in mice^{35,36} [↗](#), and its mechanism remains unknown.

Convergence index (CI), defined as the number of trans-infected cells divided by that of starter cells, was then used to estimate the tracing efficiency.^{19,32} [↗](#) We calculated the CI of traced neurons and glia separately, and found a higher tracing efficiency in larvae expressing *UGBT* than those expressing *UGTB* (**Figure 1E–G** [↗](#) and **Figure 1,2—table supplement 1** [↗](#); CI_{neuron}: 1.55 ± 0.29 vs 0.44 ± 0.11, *P* < 0.01; CI_{glia}: 4.41 ± 1.48 vs 2.50 ± 0.65, *P* = 0.30; mean ± SEM), suggesting that *UGBT* is better for G expression and tracing efficiency. Taken together, these results demonstrate that in larval zebrafish, transneuronal tracing from genetically defined cell types can be implemented by combining the injection of RVdG[EnvA] with the co-expression of helper proteins mediated by the GAL4-UAS system.

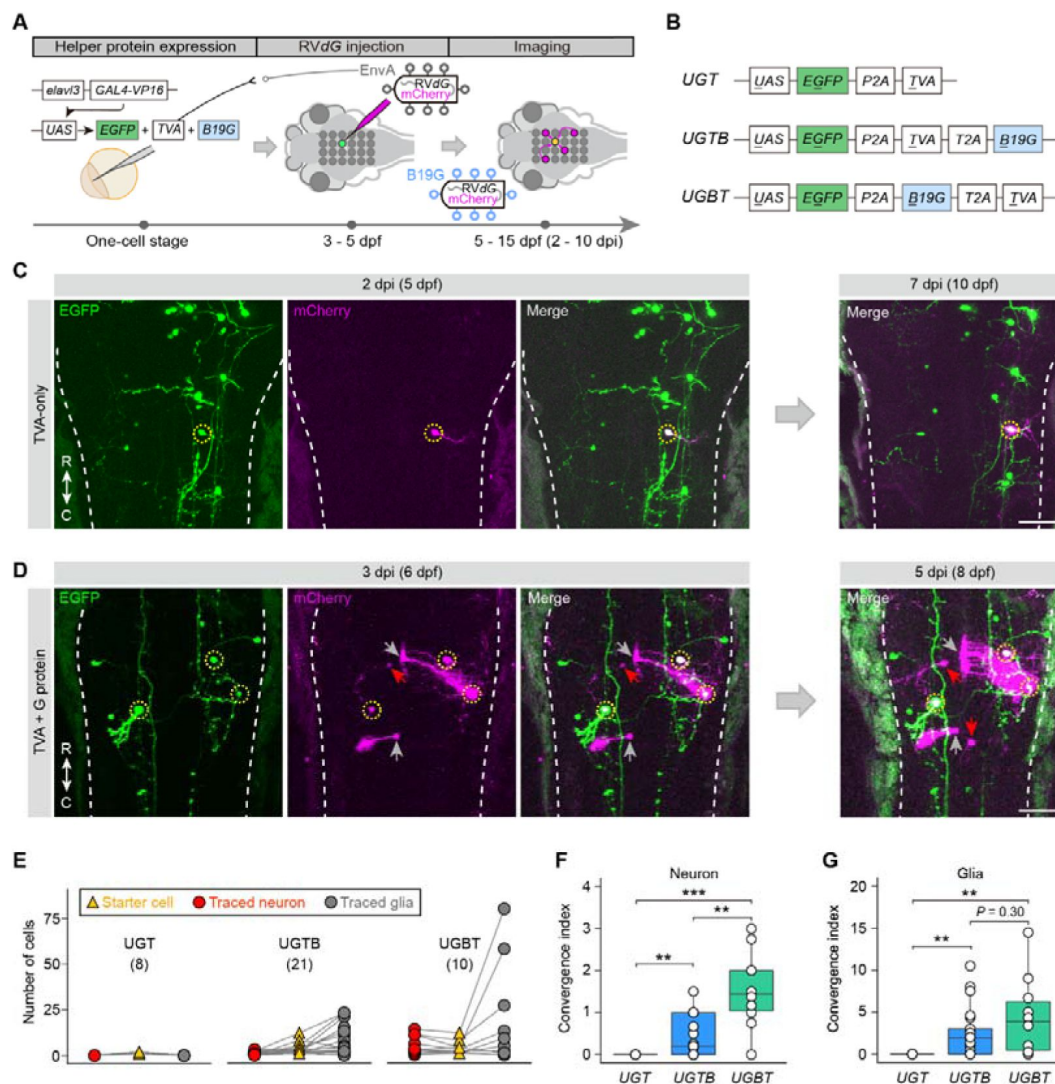


Figure 1.

Implementation of RVdG[EnvA]-based neural circuit tracing in larval zebrafish

(A) Schematic of the method used to implement RVdG[EnvA]-based circuit tracing from genetically defined cell types in larval zebrafish. Virus injections were carried out at 3–5 dpf; exact age for individual animals are listed in **Figure 1—table supplement 2**.

(B) Schematic of the three UAS-driven plasmids used to express the helper proteins TVA and G. (C and D) Time-lapse confocal images of the *nacre* larval hindbrain expressing EGFP and TVA using UGT (C), or EGFP, TVA, and B19G using UGTB (D) in randomly sparse neurons (with expression of UAS constructs driven by the *elavl3:GAL4-VP16* plasmid) and infected by SADdG-mCherry[EnvA] (magenta) injected into the hindbrain. Dashed yellow circles, starter neurons (EGFP⁺/mCherry⁺); arrows, transneuronally labeled neurons (red) and radial glia (gray) (mCherry⁺/EGFP⁺); dashed white lines, hindbrain boundaries. C, caudal; R, rostral. Scale bars, 50 μ m.

(E) Plots of the numbers of starter neurons, transneuronally traced neurons and glia in each larva expressing one of the three helper plasmids shown in (B) and injected with SADdG-mCherry[EnvA]. Data was collected between 6 and 10 dpi. Numbers in brackets indicate the number of larvae examined.

(F and G) Boxplots of the convergence index for the connection of transneuronally traced neurons

(F) and glia (G) to starter neurons shown in (E). Center, median; bounds of box, first and third quartiles; whiskers, minimum and maximum values.

****** $P < 0.01$, ******* $P < 0.001$ (nonparametric two-tailed Mann-Whitney test).

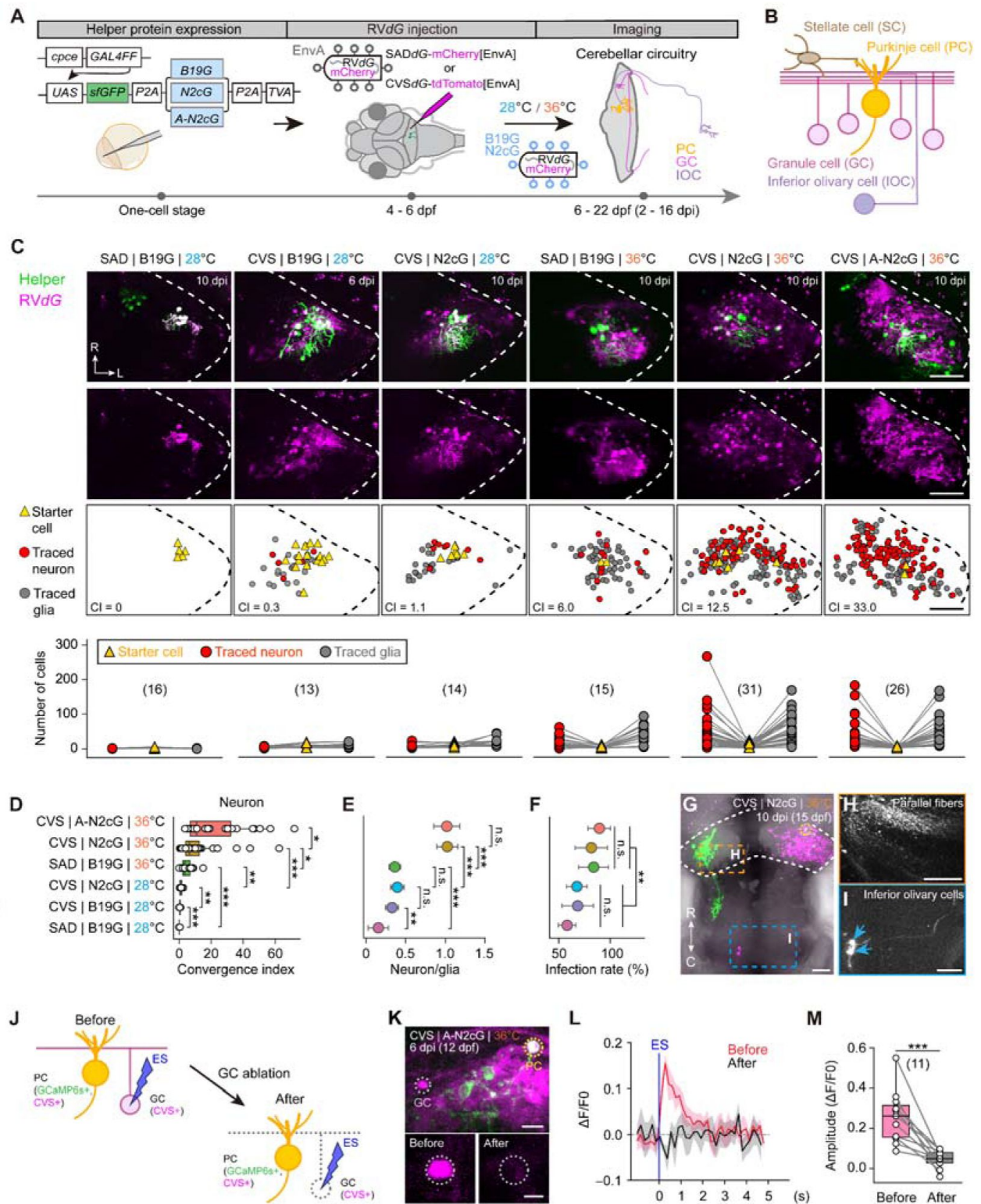


Figure 2.

Optimization and verification of RVdG[EnvA]-based retrograde monosynaptic tracing

(A) Schematic of the method used to trace inputs to PCs with RVdG[EnvA] in larval zebrafish. The *cpce:GAL4FF* plasmid and one of the three *UAS* helper plasmids were injected into *nacre* fish and combined with subsequent injection of SADdG-mCherry[EnvA] or CVSdG-tdTomato[EnvA]. Virus injections were carried out at 4–6 dpf; exact age for individual animals are listed in **Figure 1,2—table supplement 2**. The virus-injected larvae were raised at either 28 or 36°C, and the resulting labeling was monitored from 2 to 16 dpi.

(B) Schematic of the zebrafish cerebellar circuit related to inputs to PCs.

(C) Top, examples of in situ complementation in PCs under six tracing conditions. Confocal images and corresponding schematic illustrations of one cerebellar hemisphere are shown. Bottom, plots of the numbers of starter neurons, transneuronally traced neurons and glia in each larva under one of the six tracing conditions at 6–10 dpi. Dashed lines, the cerebellar boundaries. CI, convergence index of traced neurons. Numbers in brackets indicate the number of larvae examined.

(D–F) CI for the connection of transneuronally traced neurons to starter PCs (D), the ratio of the number of traced neurons to that of glia (E), and the proportion of infected fish (F, 3 to 5 independent experiments) under the six tracing conditions shown in (C, bottom).

(G), Confocal image of a larva infected with CVSdG-tdTomato[EnvA] that was trans-complemented with CVS N2cG at 36°C. Dashed yellow circle, starter PC; dashed white line, the cerebellar boundary.

(H and I) Enlarged view of the boxed regions in (G), showing transneuronally traced parallel fibers of GCs (H) and two IOCs (I). Blue arrows, somata of IOCs.

(J) Schematic of the experiment for verifying the functional connection between traced GCs and the starter PC in *nacre* larvae injected with *cpce:GAL4FF* and advanced *UG6sNT-A* plasmids and infected with CVSdG-tdTomato[EnvA] at 36°C. Ca^{2+} imaging was conducted on the starter PC in response to electrical stimulation (ES) at the GC position before and after two-photon laser ablation of the GC.

(K) Representative image (top) showing both the stimulated GC and the simultaneously recorded PC. Bottom panels display the stimulated GC before (left) and after (right) two-photon laser ablation. Dashed yellow circle, recorded PC ($\text{GCaMP6s}^+/\text{tdTomato}^+$); dashed white circles, stimulated GC ($\text{GCaMP6s}^-/\text{tdTomato}^+$).

(L) Average Ca^{2+} response (5 trials) of the same starter PC before (red) and after (black) GC ablation. The shadowed area represents SEM.

(M) Quantification of the amplitude of PC Ca^{2+} responses before and after GC ablation. The number in bracket indicate the number of GC-PC pairs examined.

Scale bars, 50 μm (C, G–I), 10 μm (K, top), 5 μm (K, bottom). SAD/CVS|B19G/N2cG/A-N2cG|28/36°C, the tracing conditions written as the G-deleted RV strain that was injected | the helper glycoprotein used for trans-complementation | fish rearing temperature after virus injection. C, caudal; L, lateral; R, rostral. Boxplots in (D and L): center, median; bounds of box, first and third quartiles; whiskers, minimum and maximum values. n.s., not-significant; $*P < 0.05$, $**P < 0.01$, $***P < 0.001$ (nonparametric two-tailed Mann-Whitney test in (D–F); two-tailed unpaired Student's *t* test in (L)). Error bars denote SEM..

Optimization and verification of the retrograde monosynaptic tracing

To optimize this system for high tracing efficiency and application feasibility in circuit research, we iterated it step-by-step in the cerebellum via testing different combinations of RV strains (SAD-B19 vs CVS-N2c), G proteins (SAD B19G vs CVS N2cG), and rearing temperatures (28°C vs 36°C) (**Figure 2A**). The cerebellar structure is evolutionarily conserved in zebrafish compared with mammals^{37,38}. In zebrafish, Purkinje cells (PCs), the central cell type in the cerebellar circuit, receive excitatory inputs from parallel fibers and climbing fibers originated respectively from granule cells (GCs) and inferior olivary cells (IOCs), and local inhibitory inputs from stellate cells (SCs)^{37,38} (**Figure 2B**). The helper proteins G and TVA were co-expressed in PCs by using the plasmid in *UGBT* mode, driven by *cpce*, a *ca8* promoter-derived PC-specific enhancer element^{37,39} (**Figure 2A**).

With the same G protein SAD B19G and rearing temperature at 28°C, CVSDG[EnvA] exhibited around a 6-fold increase in neuron tracing efficiency compared to SADdG[EnvA] (**Figure 2C,D**, **Figure 2—figure supplement 1A-C** and **Figure 1,2—table supplement 1**; CI_{neuron} : 0.41 ± 0.07 vs 0.06 ± 0.03 , $P < 0.001$). After replacing SAD B19G with CVS N2cG to complement CVSDG[EnvA], the tracing efficiency was further increased by ~ 2 times (**Figure 2C,D** and **Figure 1,2—table supplement 1**; CI_{neuron} : 1.02 ± 0.18 vs 0.41 ± 0.07 , $P < 0.01$). In addition, CVSDG[EnvA] showed a greater tendency to spread to neurons rather than glia in comparison with SADdG[EnvA] (**Figure 2C,E**). In all experiments, virus injection was performed at 4–6 days post-fertilization (dpf). The labeling of input neurons usually emerged around 2 days post injection (dpi), and all CIs were calculated at 6–10 dpi, during which the viral transfer was relatively stable (**Figure 2—figure supplement 1C**).

After elevating the temperature from 28°C to 36°C for rearing virus-injected larvae, the neuron tracing efficiency of SADdG[EnvA] became obvious (**Figure 2C,D**, **Figure 2—figure supplement 1D**, and **Figure 1,2—table supplement 1**; CI_{neuron} : 4.96 ± 1.01 vs 0.06 ± 0.03 , $P < 0.001$); for CVSDG[EnvA] trans-complemented with N2cG, the efficiency further increased by one order of magnitude (**Figure 2C,D**, **Figure 2—figure supplement 1E** and **Figure 1,2—table supplement 1**; CI_{neuron} : 11.61 ± 2.43 vs 1.02 ± 0.18 , $P < 0.001$). In addition, the elevated temperature also significantly increased the proportion of initially infected larvae (**Figure 2F**) and the tendency of viral spread to neurons (**Figure 2E**). This efficient trans-synaptic viral transfer from PCs allowed clear labeling of GC parallel fibers and long-range contralateral inputs from IOCs (**Figure 2G-I**). We then generated a *Tetoff* element-based advanced helper plasmid (*UGNT-A*) to enhance the expression level of CVS N2cG (A-N2cG) (**Figure 2—figure supplement 1F**, see Methods), and this optimization resulted in a nearly two-fold increase in the neuron tracing efficiency (**Figure 2C,D**, **Figures 2—figure supplement 1F,G** and **Figure 1,2—table supplement 1**; CI_{neuron} : 20.35 ± 3.45 vs 11.61 ± 2.43 , $P < 0.05$). The improvement was also observed for tracing glia (**Figure 2C**, **Figure 2—figure supplement 1H,I** and **Figure 1,2—table supplement 1**).

To further functionally test the synaptic specificity of RV retrograde spread in the cerebellar circuit, we conducted in vivo Ca^{2+} imaging of starter PCs while simultaneously activating single traced GCs via electrical stimulation (**Figure 2J**, see Methods). To monitor the activity of starter PCs, the fluorescent protein in the helper plasmid was replaced with the Ca^{2+} indicator GCaMP6s (*UG6sNT-A*, Methods). In total, we stimulated 33 individual GCs across 32 animals, among which 15 elicited detectable calcium responses in putative postsynaptic PCs. In 11 of these 15 cases where the stimulated GC was successfully ablated, the GC-evoked Ca^{2+} activity in starter PCs was

abolished after two-photon laser-based ablation of the GC (**Figure 2K–M**). These results indicate that RV can retrogradely spread to GCs which are functionally connected with starter PCs, confirming the capability of RV to trace monosynaptic inputs in larval zebrafish.

Time window for normal neuronal health after RV infection

In sum, we have identified the optimal conditions for efficient retrograde monosynaptic tracing in zebrafish larvae: CVSdG[EnvA] trans-complemented with A-N2cG via the GAL4-UAS system at 36°C. The high efficiency of these optimal tracing conditions prompted concerns regarding the potential cytotoxicity of CVSdG and CVS N2cG. In line with previous reports in mice²⁰, we found lower toxicity of CVSdG compared with SADdG. At 36°C, the survival rate of larvae infected with CVSdG was significantly higher than those infected with SADdG (**Figure 3A**; $75\% \pm 3\%$ vs $52\% \pm 7\%$, $P < 0.05$). Furthermore, by using the time-lapse imaging as shown in Figure S2, we often observed that early emerged starter cells disappeared in larvae infected with SADdG (**Figure 3B**). As new starter cells continued to appear from 2 to 6 dpi (**Figure 3C**), we utilized starter cells that emerged at 2 dpi to quantify their lifetime. Consistently, starter cells infected by CVSdG survived longer than those infected by SADdG (**Figure 3D**; 10.0 ± 0 days vs 7.5 ± 0.48 days, $P < 0.001$). Notably, the viability of starter cells decreased when CVSdG was trans-complemented with A-N2cG (**Figure 3D**; 10.0 ± 0 days vs 8.7 ± 0.48 days, $P < 0.001$).

To further test whether the physiological functions of starter cells were impaired by CVSdG trans-complemented with A-N2cG, we conducted in vivo whole-cell recordings on starter PCs at around 10 dpi (**Figure 3E**, see Methods). Infected PCs exhibited normal spontaneous excitatory postsynaptic currents and potentials (sEPSCs and sEPSPs), and current injection-evoked firing patterns (**Figures 3F,G**)⁴⁰. Then we examined Ca^{2+} activity of infected (i.e., starter cells) and uninfected PCs (i.e., non-starter cells) in response to complex visual stimulation⁴¹ (see Methods). No significant differences were detected between starter and non-starter PCs (**Figure 3H–L**). Taken together, these results indicate that under the optimal tracing conditions, infected neurons can remain relatively healthy up to 10 days after infection in larval zebrafish, allowing for functional probing of traced neural circuits.

Input cell-type-specific tracing and circuit reconstruction in the cerebellum

One important application of neural circuit tracing is to map the structural connectivity of specific neuron types to build the brain connectome atlas. As shown above, even under the optimal tracing conditions, glia still constituted around 47% on average of the traced cells (see **Figure 2C,E**). This will certainly affect the discrimination of traced neurons and their neural fibers. To confine retrograde tracing to neurons instead of glia, we generated a Cre-Switch⁴² transgenic reporter fish *Tg(elavl3:Tetoff-DO_DIO-Hsa.H2B-mTagBFP2_tdTomato-CAAX)*. This reporter expresses pan-neuronally nuclear-localized blue fluorescent protein (mTagBFP2) by default, which serves as a reference for image registration and data integration. It also achieves Cre-dependent neuron-specific labeling with tdTomato-CAAX via the *elavl3* promoter (**Figure 4A**, see Methods). We employed this reporter together with Cre-expressing CVSdG (CVSdG-Cre[EnvA]) to map neuronal inputs to PCs in the cerebellum (**Figure 4B**). The transgenic labeling of traced neurons showed much brighter and sharper fluorescence compared with virus-assisted labeling (**Figure 4C**). Among all 13 fish examined, we detected tdTomato-CAAX signals exclusively in neurons without any labeling of glia (**Figure 4C**), facilitating the reconstruction of traced neural circuits (**Figure 4D**). We noticed a relatively lower tracing efficiency of this Cre-dependent tracing system (**Figures 4C**, 2C; $\text{CI}_{\text{neuron}}$: 5.20 ± 1.09 vs 11.61 ± 2.43). Efforts are underway to enhance the completeness of transgene expression and improve the efficiency of Cre-mediated recombination.

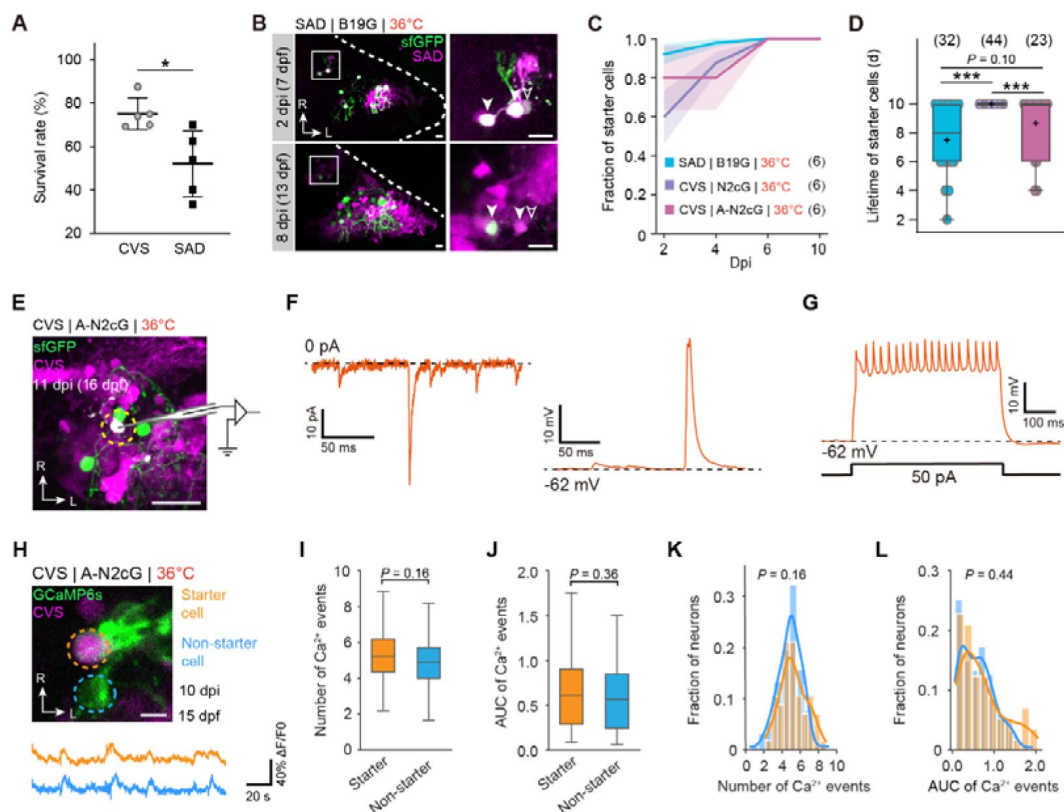


Figure 3.

Neuronal health and physiology appeared normal for at least 10 days after RV infection

(A) Survival rate of larvae injected with SADdG-mCherry[EnvA] or CVSdG-tdTomato[EnvA]. Five independent experiments were repeated.

(B) Time-lapse confocal images showing a starter cell (unfilled white arrowhead) present at 2 dpi disappears by 8 dpi, and two starter cells (white arrowheads) that remain present from 2 to 8 dpi. Boxed regions are enlarged on the right. Dashed lines indicate the cerebellar boundaries.

(C) Cumulative proportion of starter cells that have appeared at 2, 4, 6 and 10 dpi to the total number of observed starter cells under the three tracing conditions, showing the continuous appearing of starter cells at 2–6 dpi. Numbers in brackets indicate the number of larvae examined. The shadowed area represents SEM.

(D) Summary of the lifetime of starter cells appeared at 2 dpi under the three tracing conditions. The color scheme follows the same conventions shown in (C). Numbers in brackets indicate the number of starter cells examined.

(E) Schematic illustration of in vivo whole-cell recording on a starter PC at 11 dpi.

(F) Example traces of spontaneous EPSCs (left) and EPSPs (right) from starter PCs.

(G) Example trace of spikes in response to current-step injection into a starter PC.

(H) Confocal image (top) and Ca^{2+} responses (bottom) of a starter PC (dashed orange circle) and non-starter PC (dashed blue circle) to visual stimuli at 10 dpi.

(I–L) Boxplots (I and J) and distributions (K and L) of Ca^{2+} activity event number (I and K) and area under curve (AUC) of Ca^{2+} activity events (J and L) of starter PCs ($n = 88$) and non-starter PCs ($n = 127$) in larvae ($N = 8$) at 6–10 dpi.

All tracing experimental procedures were identical to those described in **Figure 2**. In panels H–L, the helper plasmid *UG6sNT-A* was used to express GCaMP6s for calcium imaging of PCs. Scale bars, 5 μm (H), 10 μm (B), 20 μm (E).

SAD/CVS|B19G/N2cG/A-N2cG|28/36°C, the tracing conditions written as the G-deleted RV strain that was injected | the helper glycoprotein used for trans-complementation | fish rearing temperature after virus injection. L, lateral; R, rostral.

Boxplots in (D, I and J): center, median; cross symbol, mean; bounds of box, first and third quartiles; whiskers, minimum and maximum values. n.s., not-significant; * $P < 0.05$; *** $P < 0.001$ (two-tailed unpaired Student's *t* test). Error bars denote SEM.

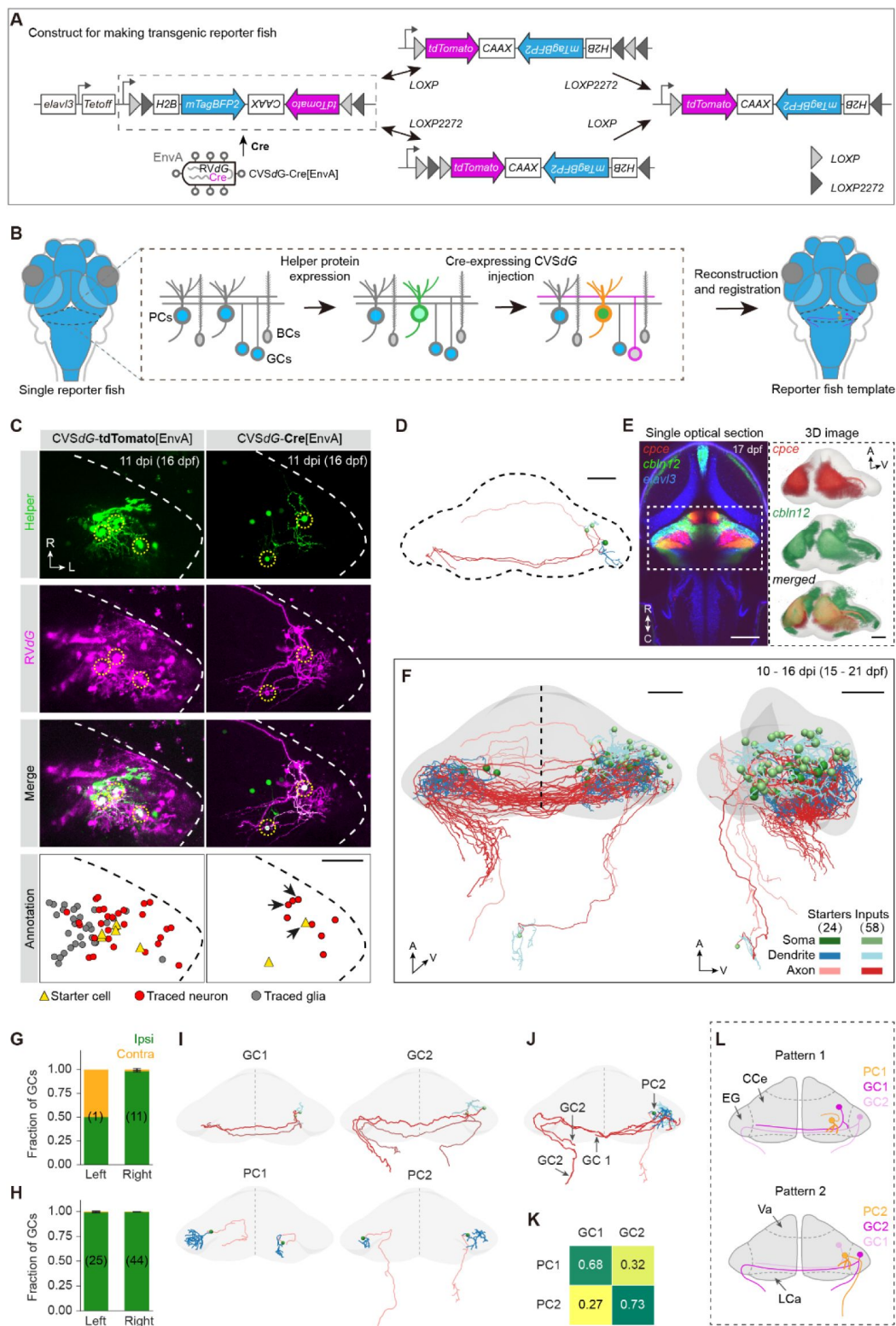


Figure 4.

Input cell-type-specific tracing and circuit reconstruction in the cerebellum

(A) A single Cre-Switch⁴² transgene (*elavl3:Tetoff-DQ_DIO-Hsa.H2B-mTagBFP2_tdTomato-CAAX*; abbr., *elavl3DoDioBR*) was generated to achieve a reference transgene (*elavl3:H2B-mTagBFP2*) for image registration and the Cre-dependent reporter expression (*tdTomato-CAAX*) specifically in neurons (*elavl3*⁺). Two ORFs flanked by two pairs of LOX sites oriented oppositely with the activation of forward-oriented *H2B-mTagBFP2* ORF by the *elavl3* promoter in global neurons. After Cre-expressing virus infection, Cre acts on either pair of LOX sites (LOXP (gray triangles) or LOXP2272 (black triangles)), resulting in stable inversion of the two ORFs and the activation of *tdTomato-CAAX* ORF by *elavl3* promoter in initially infected starter neurons and transneuronally infected input neurons.

(B) Schematic of building PC input atlas in larval zebrafish. The *elavl3DoDioBR* reporter fish (pan-neuronal H2B-mTagBFP2⁺) expressing helper proteins in PCs (sfGFP⁺/mTagBFP2⁺) were injected with Cre-expressing CVSdG[EnvA], which were trans-complemented in starter PCs and spread to direct inputs, resulting in Cre-dependent expression of the reporter protein (tdTomato-CAAX) in *elavl3*⁺ starter PCs (sfGFP⁺/tdTomato⁺/mTagBFP2⁺) and input GCs (sfGFP⁺/tdTomato⁺/mTagBFP2⁺). The pan-neuronally-expressing *H2B-mTagBFP2* in *elavl3DoDioBR* serves as the reference and bridge for the registration of reconstructed circuits. BCs, Bergmann glia; GCs, granule cells; PCs, Purkinje cells.

(C) Example of in situ complementation (11 dpi) in *nacre* larvae (left) and *elavl3DoDioBR* reporter larvae (right). Both were co-injected with *cpce-GAL4FF* and *UGNT* plasmids, and infected with CVSdG-tdTomato[EnvA] and CVSdG-Cre[EnvA], respectively. All larvae were maintained at 36°C. In the reporter larvae, Cre-dependent tdTomato expression was confined to neurons, driven by neuron-specific *elavl3* promoter. Confocal images (top) and corresponding schematic illustrations (bottom) are shown. Dashed yellow circles, starter PCs; dashed lines, the cerebellar boundaries. Black arrows indicate the reconstructed neurons shown in (D).

(D) Dorsal 3D view of the reconstructed starter PC and two input GCs. Dashed line indicates the cerebellar boundary. The color scheme follows the same conventions as in (F).

(E) Left, single horizontal section view of *Tg(2×en.cpce-E1B:tdTomato-CAAX)* (red), *Tg(cbln12:GAL4FF);Tg(5×UAS:EGFP)* (green), and *elavl3DoDioBR* (blue) templates aligned in a common coordinate space. Right, volume rendering of the cerebellar area (dashed white rectangle on the left) of the templates labeling PCs (red) and GCs (green) in outline of cerebellum region (gray).

(F) Dorsal (left) and lateral (right) 3D view of reconstructed starter PCs (n = 24) and input GCs (n = 58) in the cerebellum (gray). Different subcellular parts of PCs and GCs are color-coded. Data were collected from 13 larvae at 10–16 dpi (15–21 dpf).

(G and H) Ipsilateral vs contralateral proportions of GC inputs to PCs in the left and right cerebellum. Analysis were performed using the reconstructed data in (E) and the viral tracing data shown in **Figures 2C** and **2D** for (G) and (H), respectively. Numbers in brackets indicate the number of larvae examined which only had starter PCs on one cerebellar hemisphere.

(I) Subtypes of reconstructed PCs and GCs. Two typical cells are shown for each subtype.

(J) Example showing three GCs traced from a single PC.

(K) Summed input proportion of two subtypes of GCs to two subtypes of PCs. Cases with only one subtype of PC and clearly identifiable GCs were included. The cell numbers are as follows: PC1, 12; PC2, 6; GC1, 19; GC2, 18.

(L) Schematic representation of subtype-specific connectivity patterns from GCs to PCs. High saturation of color indicates a higher input strength. CCe, corpus cerebelli; EG, eminentia granularis; LCa, lobus caudalis cerebelli; Va, valvula cerebelli.

Scale bars, 50 μ m. A, anterior; C, caudal; L, lateral; R, rostral; V, ventral. Dashed vertical lines

indicate the midline of the cerebellum.

To demonstrate the feasibility of our tracing system in building and analyzing brain input maps, we then generated a three-dimensional (3D) reference template brain from the transgenic reporter fish at 17 dpf (see Methods). To incorporate cerebellar expression patterns, we generated the transgenic lines *Tg(2 $\times$ en.cpce:tdTomato-CAAX)* and *Tg(cbln12:GAL4FF)* to label PCs and GCs, respectively. By utilizing H2B-mTagBFP2 expression in the reporter fish as a bridge, we mapped the two cerebellar expression patterns and circuit tracing results of 13 fish mentioned above onto the common coordinate space of the reference template (**Figure 4B,E**, see Methods). We delineated the cerebellar brain area and reconstructed each starter PC ($n = 24$) and its input GCs ($n = 58$) with clearly visible morphology (**Figure 4E,F**), resulting in a digital cellular-resolution input atlas of cerebellar PCs (**Figure 4F**).

This input atlas clearly shows an ipsilateral preference of afferents from GCs to PCs (**Figure 4G**), which was in accordance with the summarized results of non-Cre-dependent viral tracing data (**Figures 4H** and **2C,D**). Interestingly, all reconstructed GCs also innervated contralaterally by crossing the midline. We found two morphological subtypes for both GCs and PCs in the reconstruction, and each included one local subtype with axonal projections within the cerebellar area and one long projection subtype with axons extending caudally to the dorsal hindbrain (**Figure 4I**). This classification of GC subtypes (GC1 and GC2) is consistent with previous studies^{37,43,44}; GC1 corresponds to rostromedial GCs with T-shaped axons and somata located in the valvula cerebelli (Va) and corpus cerebelli (CCe), whereas GC2 corresponds to caudolateral GCs that project to the dorsal hindbrain and innervate crest cells, with somata in the eminentia granularis (EG) and lobus caudalis cerebelli (LCa). Interestingly, although a single PC can receive inputs from the two subtypes of GCs (**Figure 4J**), the two subtypes of PCs prefer inputs from different subtypes of GCs (**Figure 4K,L**). These results demonstrate how cellular-level connectivity can advance our understanding of circuit wiring patterns.

Together, the development of the Cre-dependent tracing tool empowers the RVdG-based circuit tracing method, enabling input-specific tracing, cellular-level circuit reconstruction, and integrated mapping and analysis of afferent connections in larval zebrafish.

Discussion

Understanding brain functions requires systematic dissection of the involved neural circuits. Here, we developed a highly feasible and efficient implementation of the viral tracing tool RVdG[EnvA] for circuit studies in larval zebrafish (**Figure 1–4—table supplement 1**). Through step-by-step improvement, we identified an optimal tracing condition for larval zebrafish, which includes the CVSdG strain, the native G protein of the CVS (i.e., N2cG), an enhanced expression system for the G protein, and elevated rearing temperature. Our method exhibited a 20-fold increase in tracing efficiency compared to previous studies in adult zebrafish^{26,27}, and demonstrated low cytotoxicity. Moreover, we established a custom-designed transgenic reporter framework, which enables genetic access to specific input cell types and allows for cellular-resolution reconstruction and alignment of traced circuits. We demonstrated the application of this framework to decipher cell-type specific connections from GCs to PCs in the cerebellum.

Helper protein expression design that guarantees high feasibility of the tracing method

Precise spatial control of helper protein expression (i.e. TVA and G) in starter cells is critical for the application of RVdG[EnvA]-based circuit tracing. In zebrafish, previous studies expressed these proteins through generating transgenic fish lines^{26,27} or injecting helper viral vectors^{27,29,30}. Both strategies allow genetic targeting of specific neuronal types or populations via the GAL4-UAS system. However, transgenic expression typically labels the majority of the targeted neuron type, often distributed across multiple brain regions. Viral delivery can provide regional specificity, but achieving precise targeting in the compact larval zebrafish brain remains technically challenging. Our method employs the transient transgene technique in zebrafish to express helper proteins. This involves temporarily introducing GAL4-UAS binary DNA plasmids into the fish's genome through microinjection into fertilized eggs (see **Figures 1A** and **2A**). It enables sparse expression of TVA and G in genetically defined cell types, even in a single neuron. Tracing from a single neuron can help cell-subtype-specific circuit dissection.

Furthermore, we linked the two helper protein genes with the 2A element to co-express TVA and G in the same neuron (see **Figures 1A** and **2A**). Co-expression guarantees that the tracing is monosynaptically restricted. If G is expressed in neurons lacking TVA and a fluorescent indicator, these cells may act as input cells for the initially infected starter cells, resulting in further trans-complementation and spread of RVdG across the second and even more synaptic steps³². Previous studies expressed TVA and G separately, either using two transgenic lines or dual viral vectors^{26,27}, raising the risk of their expression in different neurons and potentially compromising monosynaptic specificity.

On the other hand, our helper protein expression strategy is also compatible with multi-synaptic circuit tracing. By combining sparse, cell-type-specific helper expression with an EnvA-pseudotyped, glycoprotein-competent rabies virus and time-lapse imaging, multi-step trans-synaptic spread can be monitored within individual animals. Assuming a consistent temporal delay between successive rounds of viral transmission, this approach may enable the temporal resolution of successive higher-order presynaptic populations based on their sequential emergence over time. Our study demonstrated that the transient expression of helper proteins effectively facilitates subsequent viral infection and trans-complementation, offering a time-efficient and cost-effective alternative to generating stable transgenic lines. It also addresses the potential cytotoxicity effects associated with the use of helper viral vectors. With commercially available recombinant RV tools, our method can be easily implemented by zebrafish research laboratories using three commonly used techniques: plasmid construction, microinjection (with DNA constructs and recombinant RV injected 3–6 day apart), and live optical imaging. Additional enhancements for the helper plasmid could include the implementation of light-responsive inducible systems⁴⁵ to achieve both temporal and spatial control of helper protein expression.

Key factors that influence tracing efficiency

To enhance RVdG tracing efficiency, we introduced the CVS-N2c rabies strain in zebrafish for the first time. It has been reported that the CVS-N2c strain exhibits significantly higher tracing efficiency in mice compared to the SAD-B19 strain, with an improvement of almost tenfold²⁰. In zebrafish larvae, CVSDG, when trans-complemented with its native N2cG, also showed an average increase close to tenfold compared to SADdG trans-complemented with its native B19G (see **Figure 2D** and **Figure 1,2—table supplement 1**, CI_{neuron}). This improvement may be attributed to the reduced replication and protein expression of the CVS-N2c²⁰. This could result in decreased toxicity to host cells, longer reproductive times, and ultimately higher spread efficiency. Consistently, in zebrafish larvae, we found that neurons infected with CVS-N2c displayed lower fluorescence intensity and longer survival times compared to those infected with SAD-B19.

Notably, N2cG exhibited higher tracing efficiency than B19G when they were used to trans-complement the same CVSdG (see **Figure 2D** and **Figure 1,2—table supplement 1**). This could be attributed to the greater neurotropism of N2cG compared to B19G, or potential mismatches between the cytoplasmic domain of B19G and the CVS-N2c capsid. Further investigation into the impact of different G proteins on RVdG tracing efficiency in zebrafish would benefit from the construction of chimeric G proteins for comparison⁴⁶.

Besides the virus strain, temperature also plays a pivotal role. Under the optimal temperature for virus replication at 36°C, the tracing efficiency of CVS-N2c increased more than tenfold in comparison with that under the standard rearing temperature for zebrafish at 28°C. This effect was more pronounced for SAD-B19, showing an increase of over 80 times (see **Figure 2D** and **Figure 1,2—table supplement 1**). Importantly, no significant increase in mortality or behavioral abnormalities was observed in either non-injected or virus-injected larvae maintained at elevated temperatures. In virus-injected (CVS-N2c) larvae, survival rates decreased slightly from 100% to ~80% at both 28°C and 36°C, likely due to electrode insertion-related injury rather than temperature-dependent toxicity. Virus-injected larvae could be maintained at 36°C for over a month without adverse consequences, indicating that elevated temperature is well tolerated. This is consistent with a previous report showing no temperature-dependent effects on various behaviors in adult zebrafish between 28°C and 36°C²⁷.

The expression level of G proteins is another determinant for efficient rabies tracing⁴⁶. Building upon the codon optimization of the G protein, we further increased the expression of N2cG by inserting the Tetoff element upstream of the N2cG gene (A-N2cG) in the UAS helper plasmid (see **Figure 2-figure supplement 1F**, **Figure 2D** and **Figure 1,2—table supplement 1**). This integration enables dual amplification of N2cG and TVA expression by both the GAL4-UAS and tTA-TRE binary systems. Combining the three key optimal conditions: CVS-N2c strain, 36°C temperature, and A-N2cG, we have achieved the highest tracing efficiency in zebrafish to date, with a 20-fold increase compared to previous reports²⁷.

To assess the efficacy of our trans-synaptic labeling relative to endogenous connectivity, we quantified PC, GC, and IOC populations based on transgenic labeling in *Tg(2×en.cpce-E1B:tdTomato-CAAX)*, *Tg(cbln12:GAL4FF)*; *Tg(5×UAS:EGFP)*, and *Tg(elavl3:H2B-GCaMP6s)* lines, respectively. At ~7 dpf, we counted approximately 330 PCs, 2300 GCs, and 330 IOCs in fish brain. Considering developmental scaling and anatomical constraints (e.g., a subset of GCs project exclusively ipsilaterally), we estimate that by 10–14 dpf, a single PC receives ~1000–2000 GC inputs. The IOC-PC connectivity is primarily one-to-one^{47,48}.

Under optimal tracing conditions, we labeled an average of ~20 GC inputs per PC (~1–2% capture rate) and observed IOC labeling in 7 of 248 starter PCs (~3%). While the capture efficiency is modest, it aligns with values reported in mammalian systems^{32,49}. Nevertheless, this indicates inherent limitations of rabies virus-based circuit tracing in zebrafish. Fundamentally addressing this limitation may require the development of viral tracing tools engineered from viruses that naturally infect fish⁵⁰.

Monosynaptic specificity of the tracing

Employing the cerebellar circuit as a model, we examined the complete pattern of traced neurons from PCs in larval zebrafish, and found that CVSdG was able to trace well-known presynaptic neurons of PCs, including both intra-cerebellar GCs and extra-cerebellar IOCs (see **Figure 2G-I**). Importantly, IOCs were the only labeled cells outside of the cerebellum, even after extended periods allowing for viral spread, demonstrating the retrograde and monosynaptic specificity of the rabies viral tracer in the zebrafish brain.

We further provided functional evidence that CVSdG, even at its highest efficiency with A-N2cG, spreads retrogradely between neurons through synaptic connections (see **Figure 2J–M**). In all experiments involving the successful activation of PCs in response to electrical stimulation of single GCs, we consistently did not observe PC activation after ablating the GCs. In addition, as shown in **Figure 2L**, calcium imaging at 5 Hz revealed PC responses with an average latency of 152 ± 35 ms (mean $\pm$ SEM), indicative of immediate and likely monosynaptic activation. Together with the anatomical tracing data, these findings provide strong evidence that CVSdG spreads retrogradely and monosynaptically from postsynaptic PCs to functionally connected presynaptic GCs.

Transneuronal labeling of glial cells

Transneuronal labeling of radial glia mediated by VSV, a member of the Rhabdoviridae family like RV, has been commonly observed in larval zebrafish^{29,30}. In our experiments, labeled radial glia were consistently located near the somata and dendrites of starter neurons (see **Figure 1D**). In zebrafish, radial glia—also referred to as radial astroglia—are considered functional equivalents of mammalian astrocytes. These cells are known to participate in tripartite synapses and may express endogenous receptors capable of mediating RVdG uptake at postsynaptic sites. This anatomical proximity suggests that rabies-based trans-synaptic tracing could not only resolve neuronal connectivity but also identify glial populations that are synaptically associated and functionally coupled to defined neuronal targets. Leveraging this feature offers a promising strategy for investigating neuron–glia interactions in vivo.

Glial labeling typically emerged around 2 dpi and increased over time; however, the rate of labeling plateaued or declined during the late phase (6–10 dpi), in contrast to the continued increase in neuronal labeling (see **Figure 2—figure supplement C, G**). This divergence was quantified using a labeling rate index, calculated as $R_{\text{glia}} - R_{\text{neuron}}$, which shifted from 0.87 to -0.80 between the early (2–6 dpi) and late (6–10 dpi) phases. Moreover, glial labeling observed at 2 or 4 dpi was largely diminished by 6 dpi (~ 11 dpf), regardless of onset timing. This age-dependent, rather than duration-dependent, decline is consistent with a model of abortive infection, in which the host's innate antiviral responses suppress viral replication, preventing progeny virion production and resulting in the eventual loss of reporter expression⁵¹. Supporting this interpretation, we did not observe punctate fluorescent debris characteristic of apoptotic cell death, suggesting that fluorescence loss reflects functional silencing of viral gene expression rather than glial degeneration. These findings imply that abortive infection becomes more prominent with larval maturation, potentially accounting for the negligible glial labeling observed in adult zebrafish^{26,27}.

Notably, the eventual disappearance of glial fluorescence does not indicate a lack of initial viral entry. Glial infection likely competes with neuronal infection for viral particles, thereby reducing neuronal tracing efficiency. Future studies should elucidate the molecular mechanisms underlying glial susceptibility—particularly receptor-mediated viral entry at synaptic sites. Targeted suppression of viral receptors in glial cells may help minimize off-target labeling and enhance the efficiency of neuronal circuit tracing.

Low cytotoxicity that enables functional experiments in a wide time window

We found that the CVS-N2c strain did not significantly affect the neuronal health and physiology of starter cells for at least 10 days after infection (see **Figure 3D–L**). Meanwhile, consistent with previous reports in mice²⁰, the SAD-B19 strain exhibited higher cytotoxicity than CVS-N2c, resulting in more cell death of starter cells (see **Figure 3D**). It indicates that CVS-N2c is more suitable for application in zebrafish. We could observe the labeling of starter cells even one month later after CVS-N2c infection, and the infected zebrafish could be raised to adulthood.

A potential concern is whether viral titer or injection volume contributes to cytotoxicity. However, RVdG-EnvA infection generally occurs at a rate of one viral particle per cell⁵², indicating that a higher viral dose does not proportionally increase the viral load per infected cell, and is therefore unlikely to enhance tracing efficiency or aggravate cytotoxicity at the single-cell level. Consistently, in our experiments, increasing the injection volume up to 20 nL had no detectable effect on starter cell survival.

The enhanced G protein expression achieved through the introduction of the Tetoff element (i.e. A-N2cG) in the helper plasmid improved tracing efficiency, but also resulted in elevated levels of cytotoxicity, though lower than that induced by SAD-B19 (see **Figure 3D**). This indicates an association between the expression levels of G proteins and cytotoxicity^{53,54}. Given the transferred RV is deficient in G, it is expected that retrogradely infected input cells will have lower cytotoxicity levels than starter cells. Therefore, input cells will survive for longer periods of time.

Based on our experience, the CVS-N2c can be microinjected into zebrafish larvae between 3 and 6 dpf, and stable and viable labeling of traced circuits can be observed within 6 to 16 dpi. Therefore, the observation time window ranges from 9 to 22 dpf, spanning the larval ages commonly used for whole-brain functional imaging studies^{14,55}.

Functions and expandability of the transgenic framework

In the transgenic reporter framework (i.e. *elavl3DoDioBR*) (see **Figure 4A**), we designed three expansion motifs: 1) a promoter (*elavl3*) motif that allows genetic access to the expression of reporter cassettes in specific cell types, e.g. the *vglut2a* and *gad1b* gene promoters to restrict excitatory and inhibitory targeting, respectively; 2) a Cre-dependent “Switch-On” reporter (tdTomato-CAAX) motif that allows the expression of various structural and functional tools in traced neurons after viral spread, such as photoconvertible fluorescent proteins and optogenetic tools; 3) a Cre-dependent “Switch-Off” reporter (H2B-mTagBFP2) motif that allows a default reporter to be turned off specifically in infected neurons. In addition, the helper plasmid can serve as an additional expansion motif that works together with the Cre-dependent transgenic reporter, allowing for full genetic visualization and dissection of neural circuits¹⁵.

In this study, we used the *elavl3* promoter to restrict the fluorescence reporter to neurons and the “Switch-On” reporter tdTomato-CAAX to label the membrane of traced neurons. This enables clear neuron morphology labeling (see **Figure 4C**), facilitating the reconstruction of traced neural circuits based on single-neuron morphology. We used H2B-mTagBFP2 as the default “Switch-Off” reporter to provide a reference expression pattern for registering and integrating traced circuit images from different individual fish and experiments. As an application demonstration, we applied the transgenic framework to map the monosynaptic inputs to PCs from GCs in the cerebellum of larval zebrafish at ~ 2.5 weeks old (see **Figure 4**). Interestingly, we uncovered two properties of the cerebellar circuit: 1) the ipsilateral preference of GC-to-PC connections; 2) the subtype specificity of GC-to-PC connections. Further investigation is required to elucidate the functional implications of subtype-specific GC-to-PC connections and the apparent contradiction between the bilateral projection of GC parallel fibers and the GCs’ preference for ipsilateral connections to PCs. The reference expression pattern can serve as a reference atlas to integrate other modal data from brain-wide images with information on gene expression, neurochemistry and neuronal activity. This will largely enhance our understanding of neural wiring patterns in relation to other neural characteristics⁵⁶.

Conclusion

Our study established an applicable and effective viral circuit tracing system for larval zebrafish. We believe that this tracing system will offer an avenue for the integrated structural and functional dissection of neural circuits underlying a wide range of brain functions in larval zebrafish.

Methods

Animals

Adult zebrafish (*Danio rerio*) were reared under standard laboratory conditions, with temperatures set at 28°C, and a light/dark cycle of 14 hr/10 hr. The larval zebrafish used were in *nacre* background (transparent strain with no melanophore pigmentation) and at 3–21 dpf. The sex of the larvae at this stage is not determined. All experimental procedures were approved by the Animal Care and Use Committee of the Center for Excellence in Brain Science and Intelligence Technology (CEBSIT), Chinese Academy of Sciences (NA-046-2023).

Generation of DNA constructs and transgenic lines

The GAL4-UAS binary plasmids were applied to express fluorescent protein, TVA protein, and viral G in specific neuronal types. To randomly target neurons, we used *elavl3:GAL4-VP16*⁵⁷. To target Purkinje cells (PCs), we generated a construct to drive the *GAL4FF* under the control of *cpce* (*ca8* promoter derived PC-specific enhancer element) fused to *E1B*³⁹. The *cpce* sequence was PCR-amplified from the zebrafish genome DNA, and the *cpce-E1B:GAL4FF* was flanked by a minimal *Tol2* transposase recognition sites⁵⁸. We generated seven helper plasmids (abbreviated names are in parentheses): $14 \times \text{UAS-E1B:EGFP-P2A-TVA}$ (*UGT*), $14 \times \text{UAS-E1B:EGFP-P2A-TVA-T2A-B19G}$ (*UGTB*), $14 \times \text{UAS-E1B/5} \times \text{UAS-hsp:EGFP-P2A-B19G-T2A-TVA}$ (*UGBT*), the $14 \times$ and $5 \times \text{UGBT}$ were used with *elavl3:GAL4-VP16* and *cpce-E1B:GAL4FF*, respectively), $5 \times \text{UAS-hsp:sfGFP-P2A-N2cG-P2A-TVA}$ (*UGNT*), $5 \times \text{UAS-hsp:sfGFP-P2A-Tetoff-N2cG-P2A-TVA}$ (*UGNT-A*), and $5 \times \text{UAS-hsp:GCaMP6s-P2A-Tetoff-N2cG-P2A-TVA}$ (*UG6sNT-A*). The *sfGFP*,

N2cG, *B19G*, *TVA*, and *Tetoff* were codon optimized for zebrafish and *de novo* synthesized; the *GCaMP6s* was PCR-amplified from *elavl3:GCaMP6s* plasmid (gift from M. B. Ahrens, Janelia Research Campus, USA). All these elements were sequentially assembled with 2A sequences and the linearized *miniTol2-UAS* backbone to generate polycistronic vectors through In-Fusion cloning. Sparse expression of helper proteins was achieved by the microinjection of a mixture (1 nl) of *elavl3* or *cpce*-driven GAL4 plasmid, UAS-driven helper plasmid, and *Tol2* transposase mRNA into one-cell stage zebrafish embryos at a concentration of 30 (10+10+10) ng/μl using an air-puffed pressure injector (MPPI-3, ASI).

To generate *Tg(2×en.cpce-E1B:tdTomato-CAAX)*, we created a construct to drive *tdTomato* with a 3' membrane tag encoding the CAAX box of human Harvey Ras (*CTGAACCCCTCTGATGAGAGTGGCCCCGGCTGCATGAGCTGCAAGTGTGTGCTCTCC*) under the control of two tandem *cpce* fused to E1B. To generate *Tg(cbln12:GAL4FF)* for labeling granule cells (GCs) in the cerebellum, we amplified a ~2-kbp regulatory element from zebrafish genome DNA by PCR for the *cbln12* (*cerebellin12*) gene and subcloned it into the linearized *miniTol2-GAL4FF* backbone. To generate *elavl3DoDioBR*, we first *de novo* synthesized a zebrafish codon-optimized *mTagBFP2* and constructed an intermediate plasmid containing *Tetoff-DO_DIO-Hsa.H2B-mTagBFP2_tdTomato-CAAX* using In-Fusion cloning. Subsequently, we subcloned this target fragment into the linearized *miniTol2-elavl3* backbone to generate the final vector. Transgenic zebrafish were prepared using the *Tol2* transposase-based approach. Purified target plasmid DNA was microinjected with the *Tol2* transposase mRNA into one-cell stage zebrafish embryos at a concentration of 60 (30+30) ng/μl. Injected embryos were raised to sexual maturity to identify founder fish. The *Tg(cbln12:GAL4FF)* was crossed with *Tg(5×UAS:EGFP)* to visualize GCs.

Production of EnvA-pseudotyped rabies virus

The EnvA-pseudotyped rabies viruses (RV), including SADdG-mCherry[EnvA], CVSdG-tdTomato[EnvA], and CVSdG-mCherry-2A-Cre[EnvA], were rescued and prepared following the previously reported method^{59,60}. Rabies vectors were prepared in phosphate-buffered saline (PBS) and stored at -80°C in aliquots for experimental use. The injected titers were 2×10^8 infectious units per ml (IU/ml) for SADdG-mCherry[EnvA] and CVSdG-tdTomato[EnvA], and 3×10^8 IU/ml for CVSdG-mCherry-2A-Cre[EnvA]. We found that mCherry fluorescence after CVSdG-Cre[EnvA] infection in vivo was much weaker than that of transgenetically expressed tdTomato-CAAX, and thus did not affect the observation and reconstruction of neuronal morphology.

Virus application

At 3–6 dpf, larvae expressing helper proteins in specific neurons were anesthetized using 0.03% tricaine methanesulfonate (MS-222) and mounted laterally in 1.0% low-melting agarose (Sigma) on a glass-bottom dish filled with the extracellular solution consisted of (in mM): 134 NaCl, 2.9 KCl, 2.1 $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$, 1.2 $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$, 10 HEPES, and 10 glucose (pH = 7.8, osmolality = 290 mOsm). Virus delivery was performed under a standard high-magnification upright (135 $\times$, Nikon SMZ18) or inverted fluorescence microscope (200 $\times$, Olympus IX51), enabling clear visualization of target neurons labeled with sfGFP or GCaMP6s. Approximately 2–5 nl of virus suspension was injected adjacent to the cells of interest in one hemisphere using a glass micropipette with a $\sim 10\text{ }\mu\text{m}$ tip diameter. The micropipette was held by a microinjector (Nanoliter2010, WPI) and positioned using a motorized manipulator (MC1000e, Siskiyou) to ensure accurate placement. Injection into the brain tissue but not ventricle was necessary to allow the virus solution to disperse locally and reach surrounding target neurons for efficient infection. After injection, the fish were allowed to rest for one minute before withdrawing the pipette. Then, they were placed in standard tanks in an incubator set at temperatures of 28°C or 36°C .

Confocal and two-photon imaging

Larval fish were mounted dorsal-side up in 1.5% low-melting agarose on a custom-made chamber. Structural live imaging of virus-injected fish and transgenic lines was conducted using an upright confocal laser scanning microscope (FV3000, Olympus) equipped with a 20 $\times$ water-immersion objective (N.A., 0.9). Image acquisition was performed at a lateral resolution of $0.5 \times 0.5\text{ }\mu\text{m}$ and a z-step size of $1\text{ }\mu\text{m}$. For virus-injected fish, the imaging dimension was based on the specific region of viral infection and spread. For transgenic lines, the entire brain was imaged by multiple volumes, which were then stitched together using a custom ImageJ (<http://fiji.sc>) macro based on the Grid/Collection stitching plugin⁶¹. Functional Ca^{2+} imaging of PCs expressing GCaMP6s was performed using an upright two-photon microscope (FVMPE-RS, Olympus) equipped with a 25 $\times$ water-immersion objective (N.A., 1.05) at a wavelength of 920 nm. The calcium activity was recorded at $\sim 5\text{ Hz}$ on single optical plane.

Electrical stimulation of single granule cell

Within 6 to 10 days after virus injection, larvae were paralyzed with pancuronium dibromide (PCD, 1mM, Tocris 0693) for 1 min and then mounted in 1.5% low-melting agarose on a chamber filled with the extracellular solution. The agarose and skin over the caudal side of the cerebellum were removed for the access of stimulation electrodes. Stimulation microelectrodes (1–2 μm tip opening) were created using borosilicate theta glass capillaries (BT-150-10, Sutter Instrument) through a micropipette puller (P-97, Sutter Instrument). Then a silver wire was inserted into the electrode filled with the extracellular solution. The electrode was positioned at and attached to the cell body of the target GC, which was identified by the fluorescence and morphology using confocal imaging. The stimulation was delivered by a flexible stimulus isolator (ISO-Flex, AMPI),

consisting of 5 pulse trains (6 - 8 V, 10 ms duration for one pulse, 10 pulses, 50 Hz) with inter-trial intervals of 40 s. The electrical stimulation of single GC was synchronized with the Ca^{2+} imaging of PCs by a stimulator (Master-8, AMPI).

Two-photon laser-based ablation of granule cells

A two-photon laser (795 nm) was used to ablate the GC after electrical stimulation. Confirmation of a successful ablation involved the identification of a bulb-like structure in the bright-field image and the absence of the tdTomato fluorescence signal from the targeted GC body. The efficiency of the ablation was further validated one hour later, before performing post hoc electrical stimulation at the same position using the bulb-like structure and adjacent fluorescent cells as references.

In vivo whole-cell recording of Purkinje cells

Within 6 to 15 days after virus injection, larvae were paralyzed with α -bungarotoxin (1mM) for 1 min, then mounted in 1.5% low-melting agarose on a chamber filled with the extracellular solution. The agarose and skin over the caudal side of the cerebellum were removed for the access of recording pipettes. A recording micropipette (15 - 20 M Ω , 1 - 1.5 μm tip opening) filled with internal solution was approached to the target PC by a motorized micromanipulator (MP-225, Sutter Instruments). The internal solution consisted of (in mM): 100 K-gluconate, 10 KCl, 2 $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$, 2 $\text{Mg}_2 \cdot \text{ATP}$, 0.3 $\text{GTP} \cdot \text{Na}_4$, 2 phosphocreatine, 10 EGTA, and 10 HEPES (pH = 7.4, osmolality = 270LmOsm). The target PC was identified by double fluorescence (sfGFP⁺ and tdTomato⁺) using confocal imaging and was recorded at whole-cell mode with similar procedures as our previous work⁶. The EPSP recording and current-step delivery were conducted in current-clamp mode, and the EPSC was assessed in voltage-clamp mode. Data were acquired using a patch-clamp amplifier (MultiClamp 700B, Axon Instruments) and a digitizer (Digidata 1440A, Axon Instruments), and signals were sampled at 10 kHz with Clampex 10.6 software (Molecular Devices).

Visual stimulation

Larvae examined were not anesthetized and were fully embedded in low-melting agarose. Visual stimuli were generated by a custom program written in Python and presented using an LCD screen covered with red filter paper to avoid spectrum interference. The visual stimuli basically followed a prior study⁴¹. Briefly, three trials were presented and each trial consisted of three types of stimuli in the following order: (1) four-directional whole-field gratings in black and white (including three forward, one backward, one leftward, and one rightward directions); (2) windmill patterns in black and white rotating at 0.2 Hz and varying in velocity following a sine function (including six whole, two right-half, and two left-half filed stimuli presented in alternating clockwise and counterclockwise directions); (3) whole-field flashes of 1-second duration (alternating between white and black, repeated 4 times). Except for the flash, each stimulus lasted 5 seconds, with a 5-second inter-stimulus interval.

Quantification of RVdG[EnvA]-based circuit tracing

Cells were counted by manually annotating the center of their cell bodies in image stacks using the Cell Counter plugin in ImageJ. This was done for cells showing green (sfGFP/GCaMP6s; i.e., infected and uninfected TVA-expressing cells), red (tdTomato/mCherry; i.e., traced input cells and starter cells), or yellow (both green and red; i.e., initially infected starter cells) fluorescence separately. To count the traced neurons and glia separately, we also manually identified and marked the neurons based on their significant morphological distinctions with glia in larval zebrafish. The number of traced glia was then calculated by subtracting the number of starter cells and traced neurons from the total number of cells with red fluorescence. To determine virus tracing efficiency, a convergence index (CI) was calculated as the number of traced inputs (neurons, glia, or all) divided by the number of starter cells.

Analysis of calcium imaging data

The acquired time-lapse images were aligned first, and individual regions of interest (ROIs) were manually circled on the averaged image. Each ROI corresponds to either a starter cell or a non-starter cell, and the mean grayscale value within each ROI over time was represented as F . In the electrical stimulation experiment, the baseline value F_0 was calculated as the average value of F from a 10-second interval before each stimulus. In the visual stimulation experiment, F_0 was calculated as the average value of F over a 30-second period preceding the stimuli. The relative change in the intensity of each ROI was calculated using the formula $(F - F_0)/F_0$. Response latency was defined as the time interval between the onset of electrical stimulation and the first post-stimulus imaging frame in which $\Delta F/F_0$ exceeded the pre-stimulus baseline by more than $3 \times$ standard deviations (SD). The area under the curve (AUC) and the number of Ca^{2+} activity events were determined by applying a threshold of $2 \times$ SD of a 60-second baseline activity. All of the mentioned calculations were performed using custom Python scripts.

Template generation and registration

We first generated two two-channel templates (17 dpf), *Tg(2×en.cpce:tdTomato-CAAX);elavl3DoDioBR* and *Tg(cbln12:GAL4FF);Tg(5×UAS:EGFP);elavl3DoDioBR*, by running a shell script `antsMultivariateTemplateConstruction2.sh` (`-d 3 -b 1 -g 0.1 -i 10 -c 2 -j 6 -k 2 -w 1x1 -f 12x8x4x2 -s 4x3x2x1vox -q 400x200x200x20 -r 1 -n 0 -m CC -t SyN ${image-inputs}`) on the CEBSIT's Linux computing cluster using the open-source Advanced Normalization Tools (ANTs) software⁶². Confocal images were acquired at $1 \times 1 \times 1 \mu\text{m}$ resolution. Both templates were shape-based average representations from 6 transgenic larvae after 7 iterations. Using normalized cross correlation (NCC) as a metric to evaluate the registration accuracy between *elavl3DoDioBR* templates and individual *elavl3DoDioBR* scans of larvae used for circuit tracing, we determined the *elavl3DoDioBR* in *Tg(2×en.cpce:tdTomato-CAAX);elavl3DoDioBR* template space as the common coordinate reference for registration. The expression pattern of *Tg(cbln12:GAL4FF);Tg(5×UAS:EGFP)* and circuit tracing results were transformed into the common coordinate space by using *elavl3DoDioBR* as a bridge. For image registration, we used `antsRegistration` and `antsApplyTransforms` functions in ANTs, following the registration parameters for live fish scans reported in previous work⁶³. Neuron reconstructions (SWC files⁶⁴) were aligned using `antsApplyTransformsToPoints` function in ANTs.

Cerebellum delineation and neuron reconstruction

The 3D cerebellum region was delineated manually using 3D Slicer software (<https://www.slicer.org/>) based on the expression patterns of *elavl3DoDioBR*, *Tg(2×en.cpce:tdTomato-CAAX)*, and *Tg(cbln12:GAL4FF);Tg(5×UAS:EGFP)* in the common coordinate space. The boundary delineation was performed on *elavl3DoDioBR*, assisted by overlaying the other two patterns.

To reconstruct the morphology of neurons in virus-traced circuits, neurons expressing membrane-bound tdTomato were semi-automatically traced using the simple neurite tracer (SNT)⁶⁵ plugin in ImageJ. The classification of dendritic and axonal compartments of neurites was based on structural knowledge. Neurons with low fluorescence intensity and ambiguous morphology were excluded during reconstruction. The resulting individual neuron skeletons were saved in SWC format. Tracings containing undistinguished several neuron skeletons were also saved as individual SWC files. Connectivity between individual starter PCs and traced GCs was determined by examining original confocal image stacks for clear appositions between GC axons and PC dendrites or somata.

The co-volume rendering of template cerebellum and cerebellum region (see **Figure 4E**) was performed using ParaView⁶⁶ (<https://www.paraview.org>). The co-visualization of the cerebellar region and registered neuron reconstructions (see **Figure 4F,I,J**) was performed using custom

python scripts based on NAVis package⁶⁷ (<https://github.com/navis-org/navis>).

Statistics

All statistical tests and graphs were performed with GraphPad Prism or Python software. Data normality was first examined with Shapiro-Wilk test. Comparisons between two groups with normal and non-normal distributions were made using two-tailed unpaired Student's *t* test and nonparametric two-tailed Mann-Whitney test, respectively. Differences between groups involving multiple factors were analyzed using two-way ANOVA. *P* value < 0.05 achieved statistical significance. Data are represented as mean ± SEM. The sample size and/or the number of replicates for each experiment are reported in the text, figures or figure legends.

Data availability

We have deposited the related zebrafish lines at CZRC (China Zebrafish Resource Center) and uploaded plasmid maps and sequences to Addgene. The viral vectors are available through BrainCase (Shenzhen, China). The neuron reconstructions can be downloaded at <https://doi.org/10.12412/BSDC.1748509594.10001>. All original code has been deposited at <https://github.com/soaringdu/Proj-RVCT> and is publicly available as of the date of publication. Any additional information required to reanalyze the data reported in this paper is available upon request.

Acknowledgements

We are grateful to Drs. M. B. Ahrens for sharing DNA plasmids, F. Peri and K. Kawakami for sharing zebrafish lines. We thank X.L. Shen, L. Sun, S. Li, and K. Wang for their kind help on virus injection, calcium imaging, visual stimulation, and plasmid construction experiments, respectively. This work was supported by STI2030-Major Projects (2021ZD0204500 and 2021ZD0204502), National Natural Science Foundation of China (32321003 and 62320106010), and Shanghai Municipal Science and Technology Major Project (18JC1410100 and 2018SHZDZX05). X.F.D. is also supported by the Youth Innovation Promotion Association of CAS.

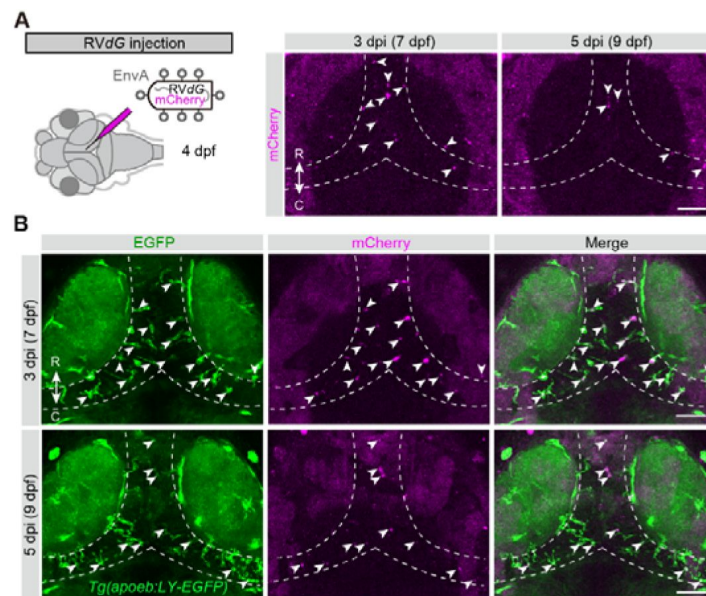


Figure 1—figure supplement 1.

No infectiousness of RVdG[EnvA] in the absence of TVA

(A) Injection of SADdG-mCherry[EnvA] into the brain of *naCre* larvae. Left, schematic; Right, time-lapse confocal images of the optic tectum of a WT larva injected with the virus.

(B) Time-lapse confocal images of the optic tectum of a *Tg(apoeb:LY-EGFP)* larva injected with SADdG-mCherry[EnvA], showing the location of mCherry-positive puncta in EGFP-positive microglia.

Scale bars, 50 μ m. Arrowheads, fluorescent-positive punctate structure with no discernible cellular morphology under high laser excitation; dashed lines, boundaries of the cell body layer of the optic tectum. C, caudal; R, rostral.

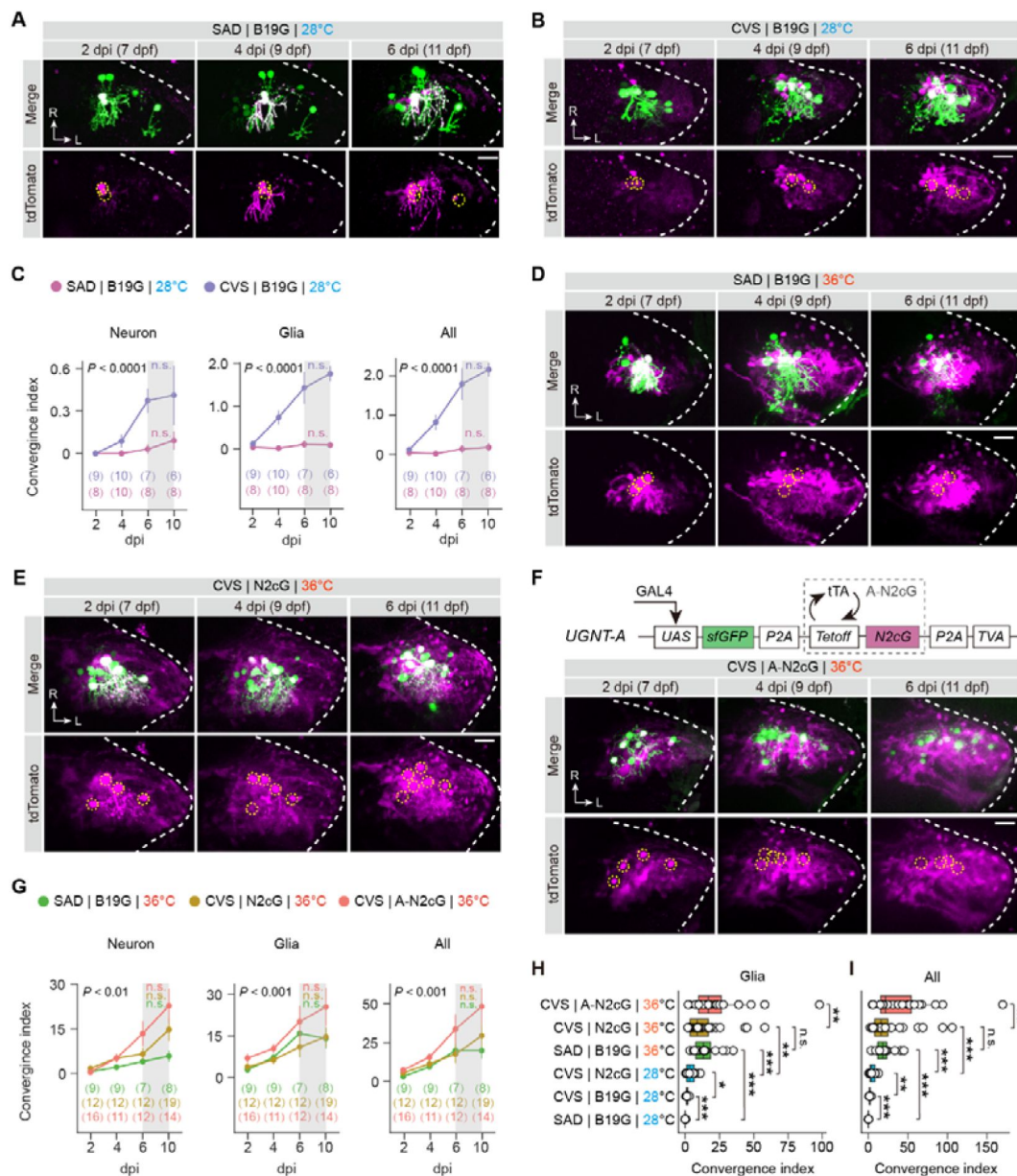


Figure 2—figure supplement 1.

Time-lapse images of trans-complemented tracing from Purkinje cells in the cerebellum under different tracing conditions

(A, B, and D–F) Time-lapse (2, 4, and 6 dpi) confocal images of in situ complementation in Purkinje cells (PCs) under tracing conditions of SAD | B19G | 28°C (A), CVS | B19G | 28°C (B), SAD | B19G | 36°C (D), CVS | N2cG | 36°C (E), and CVS | A-N2cG | 36°C (F). The schematic drawing of the advanced helper plasmid (*UGNT-A*) is shown in (F). Dashed yellow circles, starter PCs; dashed white lines, the cerebellar boundaries. L, lateral; R, rostral. Scale bars, 20 μm.

(C and G) Summary of CI for the connection of traced neurons, glia or all cells to starter PCs at 2, 4, 6 and 10 dpi under the first two tracing conditions at 28°C (C) and the last three tracing conditions at 36°C (G). Significant intergroup differences exist in CI for traced neurons, glia or all cells (see the P values). There are no statistical differences between 6 and 10 dpi in any of the groups (n.s.). Two-way ANOVA and nonparametric two-tailed Mann-Whitney test for intergroup and intragroup statistics, respectively. Error bars denote SEM. Numbers in brackets indicate the number of larvae examined.

(H and I) Boxplots of CI for the connection of traced glia (H) or all cells (I) to starter PCs under six tracing conditions at 6–10 dpi. See also **Figures 2C–E**. Center, median; bounds of box, first and third quartiles; whiskers, minimum and maximum values. n.s., not-significant; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$ (nonparametric two-tailed Mann-Whitney test).

Figure 1,2—table supplement 1.

The efficiency of viral transfer under different tracing conditions in larval zebrafish.

The convergence index (CI) was calculated as the number of traced cells divided by the number of starter cells.

Promoter	RV Strain	Helper plasmid	Glyco-protein	Temp (□)	CI _{neuron}	CI _{glia}	CI _{all}
<i>elavl3</i> (Pan-neuron)	SAD	<i>UGT</i>	No	28	0	0	0
	SAD	<i>UGTB</i>	B19G	28	0.44 ± 0.11	2.50 ± 0.65	2.94 ± 0.72
	SAD	<i>UGBT</i>	B19G	28	1.55 ± 0.29	4.41 ± 1.48	5.97 ± 1.69
<i>cpce</i> (Purkinje cell)	SAD	<i>UGBT</i>	B19G	28	0.06 ± 0.03	0.10 ± 0.05	0.16 ± 0.06
	CVS	<i>UGBT</i>	B19G	28	0.41 ± 0.07	1.58 ± 0.23	1.99 ± 0.26
	CVS	<i>UGNT</i>	N2cG	28	1.02 ± 0.18	4.20 ± 0.94	5.22 ± 1.05
	SAD	<i>UGBT</i>	B19G	36	4.96 ± 1.01	14.91 ± 2.49	19.87 ± 3.14
	CVS	<i>UGNT</i>	N2cG	36	11.61 ± 2.43	13.21 ± 2.44	24.82 ± 4.58
	CVS	<i>UGNT-A</i>	N2cG	36	20.35 ± 3.45	27.03 ± 4.37	47.38 ± 7.25
	CVS	<i>UGNT-A</i>	N2cG	36	20.35 ± 3.45	27.03 ± 4.37	47.38 ± 7.25

Figure 1,2—table supplement 2.

Injection age and corresponding fish numbers for each tracing condition.

The proportion of fish at each injection age is shown in parentheses.

Promoter	RV Strain	Helper plasmid	Glyco-protein	Temp (□)	3 dpf	4 dpf	5dpf	6 dpf
<i>elavl3</i> (Pan-neuron)	SAD	<i>UGT</i>	No	28	4 (50%)	4 (50%)	0	-
	SAD	<i>UGTB</i>	B19G	28	11 (52.4%)	9 (42.9%)	1 (4.8%)	-
	SAD	<i>UGBT</i>	B19G	28	3 (30%)	7 (70%)	0	-
<i>cpce</i> (Purkinje cell)	SAD	<i>UGBT</i>	B19G	28	-	0	16 (100%)	0
	CVS	<i>UGBT</i>	B19G	28	-	13 (100%)	0	0
	CVS	<i>UGNT</i>	N2cG	28	-	0	14 (100%)	0
	SAD	<i>UGBT</i>	B19G	36	-	0	15 (100%)	0
	CVS	<i>UGNT</i>	N2cG	36	-	0	31 (100%)	0
	CVS	<i>UGNT-A</i>	N2cG	36	-	0	23 (88.5%)	3 (11.5%)
	CVS	<i>UG6sNT-A</i>	N2cG	36	-	0	8 (25%)	24 (75%)

	This work	Dohaku <i>et al.</i> 2019 ²⁶	Satou <i>et al.</i> 2022 ²⁷		Ma <i>et al.</i> 2020 ²⁹	Kler <i>et al.</i> 2021 ³⁰
Age	3 - 21 dpf	Adult	Adult	7 - 14 dpf	2.5 - 9 dpf	3 - 9 dpf
Region being studied	Brain	Brain	Brain	Spinal cord	Brain	Brain
Virus strains	RV-SADd G[EnvA]; RV-CVSd G[EnvA]	RV-SADd G[EnvA]	RV-SADd G[EnvA]	RV-SADd G[EnvA]	rVSVd G[VSVG]	rVSV(M51R)d G[VSVG]
Glycoprotein	SAD-B19G; CVS-N2cG	SAD-B19G	SAD-B19G	SAD-B19G	VSVG	VSVG
Spread direction	Retrograde	Retrograde	Retrograde	Retrograde	Anterograde	Anterograde
Highest Tracing efficiency (CI)	~20	N.A.	~1	N.A.	N.A.	N.A.
Genetic targeting of starter cells?	Y (EnvA-TVA)	Y (EnvA-TVA)	Y (EnvA-TVA)	Y (EnvA-TVA)	N	N
Methods for helper proteins expression	Transient expression using single plasmid	Transgenesis	HSV1	HSV1	Lentivirus	Lentivirus
Co-expression of TVA & G?	Y	N	N	N	N	N
Synaptic specificity verification?	Y	N	N	N	N	N
Input or output selectivity?	Y	N	N	N	N	Y
Neuron viability test?	Y	N	Y	N	N	N
Life span of infected cells (d)	10 - 14 (starter cells)	N.A.	~6 (starter cells)	N.A.	<6 (traced cells)	<6 (traced cells)
Rearing temperature for virus-injected fish (°C)	28 & 36	~35	36	36	28	28
Experimental procedure feasibility						
1.Helper protein expression						
Plasmid microinjection						
Transgenic fish						
Helper virus						
2.Virus injection	  	   	   	   	  	  
3.Fluorescence signal amplification						
Brain slicing						
Immunohistochemistry						
4.Imaging						

Y, yes; N, no; N.A., not available

Figure 1-4—table supplement 1.

Properties of currently available viral circuit tracing systems for zebrafish.

Additional information

Author contributions

X.F.D. and J.L.D. conceptualized and supervised the project. T.L.C. and Q.S.D. developed the methodology and designed the experiments with input from X.F.D. and J.L.D.. T.L.C. and Q.S.D. performed the experiments with input from X.W. and X.D.Z.. X.D.Z. created the cerebellum-related transgenic fish. K.Z.L. produced the rabies viruses under the supervision of F.Q.X.. T.L.C. analyzed data and made the illustrations with input from Q.S.D., Y.W.Z., and X.Y.N.. Y.L. helped to establish the virus injection in zebrafish. T.L.C., Q.S.D., X.F.D., and J.L.D. wrote the manuscript.

Funding

STI2030-Major Projects (2021ZD0204500)

STI2030-Major Projects (2021ZD0204502)

National Natural Science Foundation of China (32321003)

National Natural Science Foundation of China (62320106010)

Shanghai Municipal Science and Technology Major Project (18JC1410100)

Shanghai Municipal Science and Technology Major Project (2018SHZDZX05)

Youth Innovation Promotion Association of Chinese Academy of Sciences

References

1. Raj B., Farrell J.A., Liu J., El Kholtei J., Carte A.N., Navajas Acedo J., Du L.Y., McKenna A., Relić Đ., Leslie J.M., et al. (2020) **Emergence of neuronal diversity during vertebrate brain development** *Neuron* **108**:1058–1074 [Google Scholar](#)
2. Zhang H., Wang H., Shen X., Jia X., Yu S., Qiu X., Wang Y., Du J., Yan J., He J (2021) **The landscape of regulatory genes in brain-wide neuronal phenotypes of a vertebrate brain** *eLife* **10**:e68224 <https://doi.org/10.7554/eLife.68224> | [Google Scholar](#)
3. Shainer I., Kuehn E., Laurell E., Kassar M.A., Mokayes N., Sherman S., Larsch J., Kunst M., Baier H (2023) **A single-cell resolution gene expression atlas of the larval zebrafish brain** *Sci. Adv* **9**:eade9909 [Google Scholar](#)
4. Kunst M., Laurell E., Mokayes N., Kramer A., Kubo F., Fernandes A.M., Förster D., Dal Maschio M., Baier H (2019) **A cellular-resolution atlas of the larval zebrafish brain** *Neuron* **103**:21–38 [Google Scholar](#)
5. Svara F., Förster D., Kubo F., Januszewski M., Dal Maschio M., Schubert P.J., Kornfeld J., Wanner A.A., Laurell E., Denk W., et al. (2022) **Automated synapse-level reconstruction of neural circuits in the larval zebrafish brain** *Nat. Methods* **19**:1357–1366 [Google Scholar](#)
6. Mu Y., Li X., Zhang B., Du J (2012) **Visual input modulates audiomotor function via hypothalamic dopaminergic neurons through a cooperative mechanism** *Neuron* **75**:688–699 [Google Scholar](#)
7. Yao Y., Li X., Zhang B., Yin C., Liu Y., Chen W., Zeng S., Du J (2016) **Visual cue-discriminative dopaminergic control of visuomotor transformation and behavior selection** *Neuron* **89**:598–612 [Google Scholar](#)
8. Vanwalleghe G.C., Ahrens M.B., Scott E.K (2018) **Integrative whole-brain neuroscience in larval zebrafish** *Curr. Opin. Neurobiol* **50**:136–145 [Google Scholar](#)
9. Loring M.D., Thomson E.E., Naumann E.A (2020) **Whole-brain interactions underlying zebrafish behavior** *Curr. Opin. Neurobiol* **65**:88–99 [Google Scholar](#)
10. Naumann E.A., Fitzgerald J.E., Dunn T.W., Rihel J., Sompolinsky H., Engert F (2016) **From whole-brain data to functional circuit models: the zebrafish optomotor response** *Cell* **167**:947–960 [Google Scholar](#)
11. Mu Y., Bennett D.V., Rubinov M., Narayan S., Yang C.-T., Tanimoto M., Mensh B.D., Looger L.L., Ahrens M.B (2019) **Glia Accumulate Evidence that Actions Are Futile and Suppress Unsuccessful Behavior** *Cell* **178**:27–43 [Google Scholar](#)
12. Kappel J.M., Förster D., Slangewal K., Shainer I., Svara F., Donovan J.C., Sherman S., Januszewski M., Baier H., Larsch J (2022) **Visual recognition of social signals by a tectothalamic neural circuit** *Nature* **608**:146–152 [Google Scholar](#)
13. Petrucco L., Lavian H., Wu Y.K., Svara F., Štíh V., Portugues R (2023) **Neural dynamics and architecture of the heading direction circuit in zebrafish** *Nat. Neurosci* **26**:765–773 [Google](#)

Scholar

14. Shang C.-F., Wang Y.-F., Zhao M.-T., Fan Q.-X., Zhao S., Qian Y., Xu S.-J., Mu Y., Hao J., Du J.-L. (2024) **Real-time analysis of large-scale neuronal imaging enables closed-loop investigation of neural dynamics** *Nat. Neurosci* **27**:1014–1018 [Google Scholar](#)
15. Luo L., Callaway E.M., Svoboda K. (2018) **Genetic dissection of neural circuits: a decade of progress** *Neuron* **98**:256–281 [Google Scholar](#)
16. Xu X., Holmes T.C., Luo M.-H., Beier K.T., Horwitz G.D., Zhao F., Zeng W., Hui M., Semler B.L., Sandri-Goldin R.M. (2020) **Viral vectors for neural circuit mapping and recent advances in trans-synaptic anterograde tracers** *Neuron* **107**:1029–1047 [Google Scholar](#)
17. Nectow A.R., Nestler E.J. (2020) **Viral tools for neuroscience** *Nat. Rev. Neurosci* **21**:669–681 [Google Scholar](#)
18. Wickersham I.R., Lyon D.C., Barnard R.J.O., Mori T., Finke S., Conzelmann K.-K., Young J.A.T., Callaway E.M. (2007) **Monosynaptic restriction of transsynaptic tracing from single, genetically targeted neurons** *Neuron* **53**:639–647 [Google Scholar](#)
19. Miyamichi K., Amat F., Moussavi F., Wang C., Wickersham I., Wall N.R., Taniguchi H., Tasic B., Huang Z.J., He Z., et al. (2011) **Cortical representations of olfactory input by trans-synaptic tracing** *Nature* **472**:191–196 [Google Scholar](#)
20. Reardon T.R., Murray A.J., Turi G.F., Wirblich C., Croce K.R., Schnell M.J., Jessell T.M., Losonczy A. (2016) **Rabies virus CVS-n2c ΔG strain enhances retrograde synaptic transfer and neuronal viability** *Neuron* **89**:711–724 [Google Scholar](#)
21. Barnard R.J.O., Elleder D., Young J.A.T. (2006) **Avian sarcoma and leukosis virus-receptor interactions: from classical genetics to novel insights into virus-cell membrane fusion** *Virology* **344**:25–29 [Google Scholar](#)
22. Yao S., Wang Q., Hirokawa K.E., Ouellette B., Ahmed R., Bomben J., Brouner K., Casal L., Caldejon S., Cho A., et al. (2023) **A whole-brain monosynaptic input connectome to neuron classes in mouse visual cortex** *Nat. Neurosci* **26**:350–364 [Google Scholar](#)
23. Chatterjee S., Sullivan H.A., MacLennan B.J., Xu R., Hou Y., Lavin T.K., Lea N.E., Michalski J.E., Babcock K.R., Dietrich S., et al. (2018) **Nontoxic, double-deletion-mutant rabies viral vectors for retrograde targeting of projection neurons** *Nat. Neurosci* **21**:638–646 [Google Scholar](#)
24. Jin L., Sullivan H.A., Zhu M., Lea N.E., Lavin T.K., Fu X., Matsuyama M., Hou Y., Feng G., Wickersham I.R. (2023) **Third-generation rabies viral vectors allow nontoxic retrograde targeting of projection neurons with greatly increased efficiency.** *Cell Rep Methods* **3**:100644 [Google Scholar](#)
25. Jin L., Sullivan H.A., Zhu M., Lavin T.K., Matsuyama M., Fu X., Lea N.E., Xu R., Hou Y., Rutigliani L., et al. (2024) **Long-term labeling and imaging of synaptically connected neuronal networks in vivo using double-deletion-mutant rabies viruses** *Nat. Neurosci* **27**:373–383 [Google Scholar](#)

26. Dohaku R., Yamaguchi M., Yamamoto N., Shimizu T., Osakada F., Hibi M (2019) **Tracing of afferent connections in the zebrafish cerebellum using recombinant rabies virus** *Front. Neural Circuits* **13**:30 [Google Scholar](#)
27. Satou C., Neve R.L., Oyibo H.K., Zmarz P., Huang K.-H., Arn Boudloires E., Mori T., Higashijima S., Keller G.B., Friedrich R.W (2022) **A viral toolbox for conditional and transneuronal gene expression in zebrafish** *eLife* **11**:e77153 <https://doi.org/10.7554/eLife.77153> | [Google Scholar](#)
28. Mundell N.A., Beier K.T., Pan Y.A., Lapan S.W., Göz Aytürk D., Berezovskii V.K., Wark A.R., Drokhlyansky E., Bielecki J., Born R.T., et al. (2015) **Vesicular stomatitis virus enables gene transfer and transsynaptic tracing in a wide range of organisms** *J. Comp. Neurol* **523**:1639–1663 [Google Scholar](#)
29. Ma M., Kler S., Pan Y.A (2020) **Structural neural connectivity analysis in zebrafish with restricted anterograde transneuronal viral labeling and quantitative brain mapping** *Front. Neural Circuits* **13**:85 [Google Scholar](#)
30. Kler S., Ma M., Narayan S., Ahrens M.B., Pan Y.A (2021) **Cre-dependent anterograde transsynaptic labeling and functional imaging in zebrafish using VSV with reduced cytotoxicity** *Front. Neuroanat* **15**:758350 [Google Scholar](#)
31. Federspiel M.J., Bates P., Young J.A., Varmus H.E., Hughes S.H (1994) **A system for tissue-specific gene targeting: transgenic mice susceptible to subgroup A avian leukosis virus-based retroviral vectors** *Proc. Natl. Acad. Sci* **91**:11241–11245 [Google Scholar](#)
32. Callaway E.M., Luo L (2015) **Monosynaptic circuit tracing with glycoprotein-deleted rabies viruses** *J. Neurosci* **35**:8979–8985 [Google Scholar](#)
33. Miyamichi K., Shlomei-Fuchs Y., Shu M., Weissbourd B.C., Luo L., Mizrahi A (2013) **Dissecting local circuits: parvalbumin interneurons underlie broad feedback control of olfactory bulb output** *Neuron* **80**:1232–1245 [Google Scholar](#)
34. Liu Z., Chen O., Wall J.B.J., Zheng M., Zhou Y., Wang L., Ruth Vaseghi H., Qian L., Liu J (2017) **Systematic comparison of 2A peptides for cloning multi-genes in a polycistronic vector** *Sci. Rep* **7**:2193 [Google Scholar](#)
35. Marshel J.H., Mori T., Nielsen K.J., Callaway E.M (2010) **Targeting single neuronal networks for gene expression and cell labeling in vivo** *Neuron* **67**:562–574 [Google Scholar](#)
36. Beier K.T (2021) **The Serendipity of Viral Trans-Neuronal Specificity: More Than Meets the Eye** *Front. Cell. Neurosci* **15**:720807 [Google Scholar](#)
37. Bae Y.-K., Kani S., Shimizu T., Tanabe K., Nojima H., Kimura Y., Higashijima S., Hibi M (2009) **Anatomy of zebrafish cerebellum and screen for mutations affecting its development** *Dev. Biol* **330**:406–426 [Google Scholar](#)
38. Hibi M., Shimizu T (2012) **Development of the cerebellum and cerebellar neural circuits** *Dev. Neurobiol* **72**:282–301 [Google Scholar](#)
39. Namikawa K., Dorigo A., Zagrebelsky M., Russo G., Kirmann T., Fahr W., Dübel S., Korte M., Köster R.W (2019) **Modeling neurodegenerative spinocerebellar ataxia type 13 in zebrafish**

using a purkinje neuron specific tunable coexpression system *J. Neurosci* **39**:3948–3969 [Google Scholar](#)

40. Sengupta M., Thirumalai V (2015) **AMPA receptor mediated synaptic excitation drives state-dependent bursting in purkinje neurons of zebrafish larvae** *eLife* **4**:e09158 <https://doi.org/10.7554/eLife.09158> | [Google Scholar](#)
41. Knogler L.D., Kist A.M., Portugues R (2019) **Motor context dominates output from purkinje cell functional regions during reflexive visuomotor behaviours** *eLife* **8**:e42138 <https://doi.org/10.7554/eLife.42138> | [Google Scholar](#)
42. Saunders A., Johnson C., Sabatini B (2012) **Novel recombinant adeno-associated viruses for cre activated and inactivated transgene expression in neurons** *Front. Neural Circuits* **6**:47 [Google Scholar](#)
43. Takeuchi M., Yamaguchi S., Yonemura S., Kakiguchi K., Sato Y., Higashiyama T., Shimizu T., Hibi M (2015) **Type IV collagen controls the axogenesis of cerebellar granule cells by regulating basement membrane integrity in zebrafish** *PLOS Genet* **11**:e1005587 [Google Scholar](#)
44. Matsui H., Namikawa K., Babaryka A., Koster R.W (2014) **Functional regionalization of the teleost cerebellum analyzed in vivo** *Proc. Natl. Acad. Sci* **111**:11846–11851 [Google Scholar](#)
45. Morikawa K., Furuhashi K., de Sena-Tomas C., Garcia-Garcia A.L., Bekdash R., Klein A.D., Gallerani N., Yamamoto H.E., Park S.-H.E., Collins G.S., et al. (2020) **Photoactivatable Cre recombinase 3.0 for in vivo mouse applications** *Nat. Commun* **11**:2141 [Google Scholar](#)
46. Kim E.J., Jacobs M.W., Ito-Cole T., Callaway E.M (2016) **Improved Monosynaptic Neural Circuit Tracing Using Engineered Rabies Virus Glycoproteins** *Cell Rep* **15**:692–699 [Google Scholar](#)
47. Hsieh J.-Y., Ulrich B., Issa F.A., Wan J., Papazian D.M (2014) **Rapid development of Purkinje cell excitability, functional cerebellar circuit, and afferent sensory input to cerebellum in zebrafish** *Front. Neural Circuits* **8** [Google Scholar](#)
48. Hamling K., Tobias Z., Weissman T (2015) **Mapping the development of cerebellar Purkinje cells in zebrafish** *Dev. Neurobiol* **75** [Google Scholar](#)
49. Wall N.R., Wickersham I.R., Cetin A., De La Parra M., Callaway E.M. (2010) **Monosynaptic circuit tracing in vivo through Cre-dependent targeting and complementation of modified rabies virus** *Proc. Natl. Acad. Sci* **107**:21848–21853 [Google Scholar](#)
50. Lei Y., Xiong Y., Tao D., Wang T., Chen T., Du X., Cao G., Tu J., Dai J (2022) **Construction of attenuated strains for red-spotted grouper nervous necrosis virus (RGNNV) via reverse genetic system** *Viruses* **14**:1737 [Google Scholar](#)
51. Tian B., Zhou M., Yang Y., Yu L., Luo Z., Tian D., Wang K., Cui M., Chen H., Fu Z.F., et al. (2018) **Lab-Attenuated Rabies Virus Causes Abortive Infection and Induces Cytokine Expression in Astrocytes by Activating Mitochondrial Antiviral-Signaling Protein Signaling Pathway** *Front. Immunol* **8**:2011 [Google Scholar](#)

52. Zhang A., Jin L., Yao S., Matsuyama M., van Velthoven C.T., Sullivan H.A., Sun N., Kellis M., Tasic B., Wickersham I., et al. (2024) **Rabies virus-based barcoded neuroanatomy resolved by single-cell RNA and in situ sequencing** *eLife* **12**:RP87866 [Google Scholar](#)
53. Faber M., Pulmanusahakul R., Hodawadekar S.S., Spitsin S., McGettigan J.P., Schnell M.J., Dietzschold B (2002) **Overexpression of the rabies virus glycoprotein results in enhancement of apoptosis and antiviral immune response** *J. Virol* **76**:3374–3381 [Google Scholar](#)
54. Ohara S., Sato S., Oyama K., Tsutsui K.-I., Iijima T (2013) **Rabies virus vector transgene expression level and cytotoxicity improvement induced by deletion of glycoprotein gene** *PLoS One* **8**:e80245 [Google Scholar](#)
55. Cong L., Wang Z., Chai Y., Hang W., Shang C., Yang W., Bai L., Du J., Wang K., Wen Q (2017) **Rapid whole brain imaging of neural activity in freely behaving larval zebrafish (danio rerio)** *eLife* **6**:e28158 <https://doi.org/10.7554/eLife.28158> | [Google Scholar](#)
56. Leergaard T.B., Bjaalie J.G (2022) **Atlas-based data integration for mapping the connections and architecture of the brain** *Science* **378**:488–492 [Google Scholar](#)
57. Du X., Xu B., Zhang Y., Chen M., Du J (2018) **A transgenic zebrafish model for in vivo long-term imaging of retinotectal synaptogenesis** *Sci. Rep* **8**:14077 [Google Scholar](#)
58. Kawakami K (2005) **Transposon tools and methods in zebrafish** *Dev. Dyn* **234**:244–254 [Google Scholar](#)
59. Osakada F., Callaway E.M (2013) **Design and generation of recombinant rabies virus vectors** *Nat. Protoc* **8**:1583–1601 [Google Scholar](#)
60. Lin K.-Z., Li L., Ma W.-Y., Yang X., Han Z.-P., Luo N.-S., Wang J., Xu F.-Q (2022) **A rabies virus-based toolkit for efficient retrograde labeling and monosynaptic tracing** *Neural Regen Res* **18**:1827–1833 [Google Scholar](#)
61. Preibisch S., Saalfeld S., Tomancak P (2009) **Globally optimal stitching of tiled 3D microscopic image acquisitions** *Bioinformatics* **25**:1463–1465 [Google Scholar](#)
62. Avants B.B., Yushkevich P., Pluta J., Minkoff D., Korczykowski M., Detre J., Gee J.C (2010) **The optimal template effect in hippocampus studies of diseased populations** *Neuroimage* **49**:2457–2466 [Google Scholar](#)
63. Marquart G.D., Tabor K.M., Horstick E.J., Brown M., Geoca A.K., Polys N.F., Nogare D.D., Burgess H.A (2017) **High-precision registration between zebrafish brain atlases using symmetric diffeomorphic normalization** *GigaScience* **6** [Google Scholar](#)
64. Cannon R.C., Turner D.A., Pyapali G.K., Wheal H.V (1998) **An on-line archive of reconstructed hippocampal neurons** *J. Neurosci. Methods* **84**:49–54 [Google Scholar](#)
65. Arshadi C., Günther U., Eddison M., Harrington K.I.S., Ferreira T.A (2021) **SNT: a unifying toolbox for quantification of neuronal anatomy** *Nat. Methods* **18**:374–377 [Google Scholar](#)
66. Ahrens J., Geveci B., Law C (2005) **ParaView: An End-User Tool for Large-Data Visualization** *Visualization Handbook* :717–731 [Google Scholar](#)

67. Schlegel P., Barnes C., Champion A., Jagannathan S., Court R., Pedigo B., Newstein P., Azatian Y, et al (2023) **Navis-org/navis: version 1.5.0** *Zenodo*

Author information

Tian-Lun Chen[†]

Institute of Neuroscience, State Key Laboratory of Neuroscience, Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, China, University of Chinese Academy of Sciences, Beijing, China
ORCID iD: [0000-0002-3802-955X](https://orcid.org/0000-0002-3802-955X)

[†]These authors contributed equally

Qiu-Sui Deng[†]

Institute of Neuroscience, State Key Laboratory of Neuroscience, Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, China, University of Chinese Academy of Sciences, Beijing, China

[†]These authors contributed equally

Kun-Zhang Lin

The Brain Cognition and Brain Disease Institute (BCBDI), Shenzhen Key Laboratory of Viral Vectors for Biomedicine, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China

Xiu-Dan Zheng

Institute of Neuroscience, State Key Laboratory of Neuroscience, Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, China

Xin Wang

Institute of Neuroscience, State Key Laboratory of Neuroscience, Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, China, University of Chinese Academy of Sciences, Beijing, China

Yong-Wei Zhong

Institute of Neuroscience, State Key Laboratory of Neuroscience, Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, China

Xin-Yu Ning

Institute of Neuroscience, State Key Laboratory of Neuroscience, Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, China

Ying Li

Institute of Neuroscience, State Key Laboratory of Neuroscience, Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, China

Fu-Qiang Xu

The Brain Cognition and Brain Disease Institute (BCBDI), Shenzhen Key Laboratory of Viral Vectors for Biomedicine, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China

Jiu-Lin Du

Institute of Neuroscience, State Key Laboratory of Neuroscience, Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, China, University of Chinese Academy of Sciences, Beijing, China, School of Life Science and Technology, ShanghaiTech University, Shanghai, China

For correspondence: forestdu@ion.ac.cn

Xu-Fei Du

Institute of Neuroscience, State Key Laboratory of Neuroscience, Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, Shanghai, China, University of Chinese Academy of Sciences, Beijing, China
ORCID iD: [0000-0002-4599-0074](https://orcid.org/0000-0002-4599-0074)

For correspondence: xufeidu@ion.ac.cn

Editors

Reviewing Editor

Xin Duan

University of California, San Francisco, San Francisco, United States of America

Senior Editor

Albert Cardona

University of Cambridge, Cambridge, United Kingdom

Reviewer #1 (Public review):**(1) Presentation of Figures in the Response Letter**

I would like to note that the figures included in the response letter would benefit from improved organization. For example, Author response image 1 lacks clarity for experimental conditions. From the response letter, my understanding is that a "Labeling rate index", $R_g - R_n$, was calculated to represent the difference in the rate of increase in labeling between neurons and glial across two time intervals based on experiments shown in Figure 2-figure supplement 1C and G. It seems that a mean convergence index was calculated for each experimental condition at each time point for glial and neurons, and then the differences in mean convergence index increase between time intervals were calculated for glial and neurons. The legend needs more detail to enhance clarity.

Furthermore, the manuscript should clearly distinguish between figures generated from re-analysis of existing data and those based on newly conducted experiments. This distinction should be explicitly stated in the figure legends and/or main text.

I recommend that all response figures containing data integral to the authors' rebuttal be properly integrated into the manuscript's existing supplementary figure set, rather than remaining isolated in the response document. This would enhance clarity and ensure that key supporting data are fully accessible to readers. For instance, Author response image 1 can be integrated with Figure 2-figure supplement.

(2) Glial Cell Labeling and Specificity of Trans-Synaptic Spread

The authors provided a comprehensive and well-reasoned response to the concern regarding the labeling of radial glial cells. The inclusion of a dedicated section in the revised Discussion and response figures (possibly to be integrated with supplementary figures), strengthens the manuscript.

The authors have made an interesting observation in Author response image 2 that glial labeling was frequently observed near the soma and dendrites of starter cells, suggesting that transneuronal labeled glial cells may be synaptically associated with the starter neurons. Also astroglia starter cells lead to infection of nearby TVA-negative astroglia, suggesting astroglia-to-astroglia transmission.

I find the response scientifically satisfactory and appreciate the authors' transparency in addressing the limitations of their approach.

(3) Temperature Effects and Larval Viability

The authors' justification for raising larvae at 36C to improve labeling efficiency is reasonable. The supporting data indicating minimal impact on larval viability within the experimental timeframe are convincing. Referencing prior behavioral studies and including survival data under controlled conditions adds credibility to their claims. I find this issue satisfactorily addressed.

(4) Viral Toxicity and Dosage Considerations, Secondary Starter Cells

The authors present a well-reasoned explanation that viral cytotoxicity is primarily driven by replication and not by viral titer or injection volume. However, the inclusion of experimental data directly testing the effects of higher titer or volume on starter cell viability would have strengthened this point, particularly since such tests are relatively straightforward to perform.

Regarding the potential contribution of secondary starter cells, the authors provide a convincing rationale for why such effects are unlikely under their sparse labeling conditions. However, in cases where TVA and G are broadly expressed-such as under the vglut2a promoter, as shown in Author response image 2-it would be valuable to directly evaluate this possibility experimentally. While the authors' interpretation is reasonable, empirical validation would further strengthen their conclusions.

<https://doi.org/10.7554/eLife.100880.2.sa2>

Reviewer #2 (Public review):

The study by Chen, Deng et al. aims to develop an efficient viral transneuronal tracing method that allows efficient retrograde tracing in the larval zebrafish. The authors utilize pseudotyped-rabies virus that can be targeted to specific cell types using the EnvA-TvA systems. Pseudotyped rabies virus has been used extensively in rodent models and, in recent years, has begun to be developed for use in adult zebrafish. However, compared to rodents, the efficiency of spread in adult zebrafish is very low (~one upstream neuron labeled per starter cell). Additionally, there is limited evidence of retrograde tracing with pseudotyped rabies in the larval stage, which is the stage when most functional neural imaging studies are done in the field. In this study, the authors systematically optimized several parameters of rabies tracing, including different rabies virus strains, glycoprotein types, temperatures, expression construct designs, and elimination of glial labeling. The optimal configurations developed by the authors are up to 5-10 fold higher than more typically used configurations.

The results are convincing and support the conclusions. There are some additional changes that are recommended:

(1) The new data included in the response to reviewer's letter are important to support the main conclusions and should be included in the manuscript.

(2) Line 357-362: This section should include all of the Author response image and associated details. Additionally, the Author response image 3 is at odds with Fig 2-supplement 1G. In Author response image 3, ~75% of glial cells labeled at 4 dpi loses their fluorescence by 10 dpi. However, Figure 2-supplement 1G shows that glial overall labeling increases ~2 fold from 4 dpi to 10 dpi. This would suggest that the de novo labeling rate for glia is much higher than the net labeling rate calculated from the convergence index. The authors should clarify these findings.

<https://doi.org/10.7554/eLife.100880.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

EnvA-pseudotyped glycoprotein-deleted rabies virus has emerged as an essential tool for tracing monosynaptic inputs to genetically defined neuron populations in the mammalian brain. Recently, in addition to the SAD B19 rabies virus strain first described by Callaway and colleagues in 2007, the CVS N2c rabies virus strain has become popular due to its low toxicity and high trans-synaptic transfer efficiency. However, despite its widespread use in the mammalian brain, particularly in mice, the application of this cell-type-specific monosynaptic rabies tracing system in zebrafish has been limited by low labeling efficiency and high toxicity. In this manuscript, the authors aimed to develop an efficient retrograde monosynaptic rabies-mediated circuit mapping tool for larval zebrafish. Given the translucent nature of larval zebrafish, whole-brain neuronal activities can be monitored, perturbed, and recorded over time. Introducing a robust circuit mapping tool for larval zebrafish would enable researchers to simultaneously investigate the structure and function of neural circuits, which would be of significant interest to the neural circuit research community. Furthermore, the ability to track rabies-labeled cells over time in the transparent brain could enhance our understanding of the trans-synaptic retrograde tracing mechanism of the rabies virus.

To establish an efficient rabies virus tracing system in the larval zebrafish brain, the authors conducted meticulous side-by-side experiments to determine the optimal combination of trans-expressed rabies G proteins, TVA receptors, and recombinant rabies virus strains. Consistent with observations in the mouse brain, the CVS N2c strain trans-complemented with N2cG was found to be superior to the SAD B19 combination, offering lower toxicity and higher efficiency in labeling presynaptic neurons. Additionally, the authors tested various temperatures for the larvae post-virus injection and identified 36 °C as the optimal temperature for improved virus labeling. They then validated the system in the cerebellar circuits, noting evolutionary conservation in the cerebellar structure between zebrafish and mammals. The monosynaptic inputs to Purkinje cells from granule cells were neatly confirmed through ablation experiments.

However, there are a couple of issues that this study should address. Additionally, conducting some extra experiments could provide valuable information to the broader research field utilizing recombinant rabies viruses as retrograde tracers.

(1) It was observed that many radial glia were labeled, which casts doubt on the specificity of trans-synaptic spread between neurons. The issues of transneuronal labeling of glial cells should be addressed and discussed in more detail. In this manuscript, the authors used a transgenic zebrafish line carrying a neuron-specific Cre-dependent reporter and EnvA-CVS N2c(dG)-Cre virus to avoid the visualization of virally infected glial cells. However, this does not solve the real issue of glial cell labeling and the possibility of a nonsynaptic spread mechanism.

In agreement with the reviewer's suggestion, we have incorporated a standalone section in the revised Discussion (page 9) to address the issue of transneuronal glial labeling, including its spatial distribution, temporal dynamics, potential mechanisms, and possible strategies for real resolution.

Regarding the specificity of trans-synaptic spread between neurons, we have demonstrated that our transsynaptic tracing system reliably and specifically labels input neurons. Structurally, we only observed labeling of inferior olivary cells (IOCs) outside the cerebellum, which are the only known extracerebellar inputs to Purkinje cells (PCs), while all other traced neurons remained confined within the cerebellum throughout the observation period (see Figure 2G–I). Functionally, we verified that the traced neurons formed synaptic connections with the starter PCs (see Figure 2J–M). Together, these findings support the conclusion that our system enables robust and specific retrograde monosynaptic tracing of neurons in larval zebrafish.

Regarding the transneuronal labeling of radial glia cells, we observed that their distribution closely correlates with the location of neuronal somata and dendrites (see Author response image 2). In zebrafish, radial glial cells are considered functional analogs of astrocytes and are often referred to as radial astroglia. The adjacent labeled astroglia may participate in tripartite synapses with the starter neurons and express viral receptors that enable RV particle entry at postsynaptic sites. This suggests that rabies-based tracing in zebrafish may serve as a valuable tool for identifying synaptically associated and functionally connected glia. Leveraging this approach to investigate glia–neuron interactions represents a promising direction for future research.

In our system, the glial labeling diminishes at later larval stages, likely due to abortive infection (see Author response image 3 and relevant response). However, the eventual clearance of infection does not preclude the initial infection of glial cells, which may compete with neuronal labeling and reduce overall tracing efficiency. Notably, transneuronal infection of glial cells by RV has also been observed in mammals (Marshel et al., 2010). To minimize such off-target labeling, future work should focus on elucidating the mechanisms underlying glial susceptibility—such as receptor-mediated viral entry— and developing strategies to suppress receptor expression specifically in glia, thereby improving the specificity and efficiency of neuronal circuit tracing.

In addition, wrong citations in Line 307 were made when referring to previous studies discovering the same issue of RVdG-based transneuronal labeling radial glial cells. "The RVdG-based transneuronal labeling of radial glial cells was commonly observed in larval zebrafish29,30".

The cited work was conducted using vesicular stomatitis virus (VSV). A more thorough analysis and/or discussion on this topic should be included.

We thank the reviewer for pointing out the citation inaccuracy. The referenced study employed vesicular stomatitis virus (VSV), which, like RV, is a member of the Rhabdoviridae family. We have revised the text accordingly—from "RVdG-based transneuronal labeling of radial glial cells..." to "Transneuronal labeling of radial glial cells mediated by VSV, a member

of the Rhabdoviridae family like RV, has been commonly observed in larval zebrafish" (page 9, line 347).

Several key questions should be addressed:

Does the number of labeled glial cells increase over time?

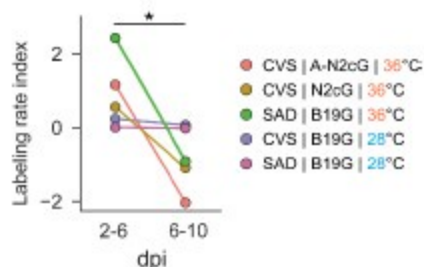
Yes, as shown in Figure 2—figure supplement 1C and G, the number of labeled radial glial cells significantly increased from 2 to 6 days post-injection (dpi). This phenomenon has been addressed in the revised Discussion section (page 9, line 357).

Do they increase at the same rate over time as labeled neurons?

Although glial cell labeling continued to increase over time, we observed a slowdown in labeling rate between 6 and 10 dpi, as shown in Figure 2—figure supplement 1C and G. Therefore, we divided the timeline into two intervals (2–6 and 6–10 dpi) to compare the rate of increase in labeling between neurons and glia. The rate (R) was defined as the daily change in convergence index. To quantify the difference between neuronal and glial labeling rates, we calculated a labeling rate index: $R_g - R_n$, where R_g and R_n denote the rates for glia and neurons, respectively (Author response image1). Our analysis revealed that, between 2 and 6 dpi, glial cells exhibited a higher labeling rate than neurons. However, this trend reversed between 6 and 10 dpi, with neurons surpassing glial cells in labeling rate. These findings have been included in the revised Discussion section (page 9).

Author response image 1.

Labeling rate index of glia and neurons across two time intervals. Data points represent the mean labeling rate index for each tracing strategy within each time interval. *P < 0.05 (nonparametric two-tailed Mann-Whitney test).



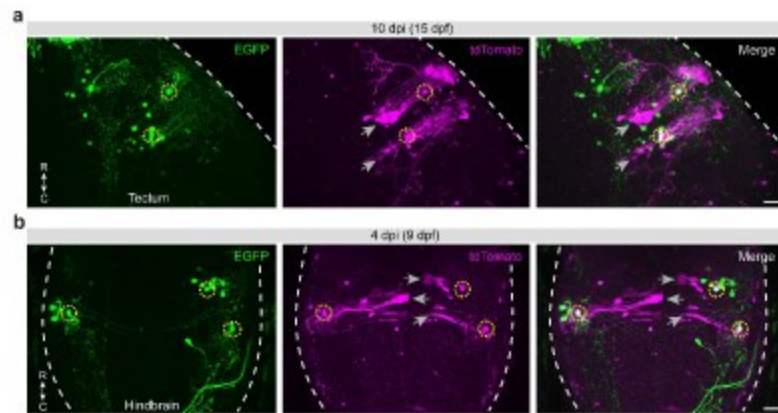
Are the labeled glial cells only present around the injection site?

We believe the reviewer is inquiring whether labeled glial cells are spatially restricted to the vicinity of starter neurons. The initial infection is determined by the expression of TVA rather than the injection site. For example, injecting a high volume of virus into the anterior hindbrain resulted in the infection of TVA-expressing cells in distant regions, including the 109 tectum and posterior hindbrain (Author response image 2).

Regarding glial labeling, PC starter experiments showed that labeled glial cells (i.e. Bergmann glia) were predominantly localized within the cerebellum, likely due to the confinement of PC dendrites to this region. When using vglut2a to define starter neurons, glial labeling was frequently observed near the soma and dendrites of starter cells (14 out of 114 of 17 cases; Author response image 2). These observations suggest that transneuronal labeled glial cells may be synaptically associated with the starter neurons. We have included this point in the revised Discussion section (page 9).

Author response image 2.

Location of transneuronal labeled glial cells. (a and b) Confocal images showing the right tectum (a) and posterior hindbrain (b) of different WT larvae expressing EGFP and TVA using UGNT in randomly sparse neurons ($vglut2a^+$) and infected with CVSdGtdTomato[EnvA] (magenta) injected into the anterior hindbrain. Dashed yellow circles, starter neurons ($EGFP^+/tdTomato^+$); gray arrows, transneuronally labeled radial glia ($tdTomato^+/EGFP^-$); dashed white lines, tectum or hindbrain boundaries. C, caudal; R, rostral. Scale bars, 20 μm .



Can the phenomenon of transneuronal labeling of radial glial cells be mitigated if the tracing is done in slightly older larvae?

Yes, we agree. As elaborated in the following response, we hypothesize that the loss of fluorescence in radial glial cells at later developmental stages is due to abortive infection (see Author response image 3 and associated response). This supports the notion that abortive infection becomes increasingly pronounced as larvae mature, potentially explaining the negligible glial labeling observed in adult zebrafish (Dohaku et al., 2019; Satou et al., 2022). However, as noted in our response to the first comment, the disappearance of fluorescence does not indicate the absence of viral entry. Viral receptors may express on glial cells, allowing initial infection despite a failure in subsequent replication. Consequently, glial infection—though abortive—may still compete with neuronal infection and reduce tracing efficiency.

What is the survival rate of the infected glial cells over time?

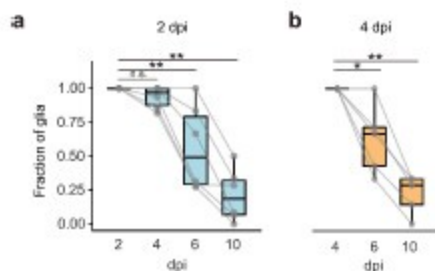
We observed the disappearance of glial fluorescence after transneuronal labeling, while we did not observe punctate fluorescent debris typically indicative of apoptotic cell death. Therefore, we favor the hypothesis that the loss of glial fluorescence results from abortive infection rather than cell death. Abortive infection refers to a scenario in which viral replication is actively suppressed by host antiviral responses, preventing the production of infectious viral particles. For example, recent studies have shown that lab-attenuated rabies virus (RV) induces the accumulation of aberrant double-stranded DNA in astrocytes, which activates mitochondrial antiviral-signaling protein (MAVS) and subsequent interferon expression (Tian et al., 2018). This antiviral response inhibits RV replication, ultimately resulting in abortive infection.

In addition, we quantified the proportion of glial cells labeled at 2 dpi and 4dpi that retained fluorescence over time. By 6 dpi (approximately 11 dpf), glial labeling had largely diminished in both groups (Author response image 3). These results suggest that the decline in glial

fluorescence is more closely linked to larval age than to the duration of glial infection, supporting the notion of abortive infection. This also addresses the reviewer's earlier concern and indicates that glial labeling is mitigated in older larvae.

Author response image 3.

Fraction of glial cells with fluorescence retention. (a and b) Proportion of glial cells labeled at 2 dpi (a) and 4 dpi (b) that retained fluorescence over time. Data are from the CVS|N2cG|36°C group. In boxplots: center, median; bounds of box, first and third quartiles; whiskers, minimum and maximum values. n.s., not-significant; * $P < 0.05$, ** $P < 0.01$ (nonparametric two-tailed Mann-Whitney test).



If an infected glial cell dies due to infection or gets ablated, does the rabies virus spread from the dead glial cells?

In our system, glial cells do not express the rabies glycoprotein (G). Therefore, even if glial cells are transneuronally infected, they cannot support viral budding or assembly of infectious particles due to the absence of G (Mebatsion et al., 1996), preventing further viral propagation to neighboring cells.

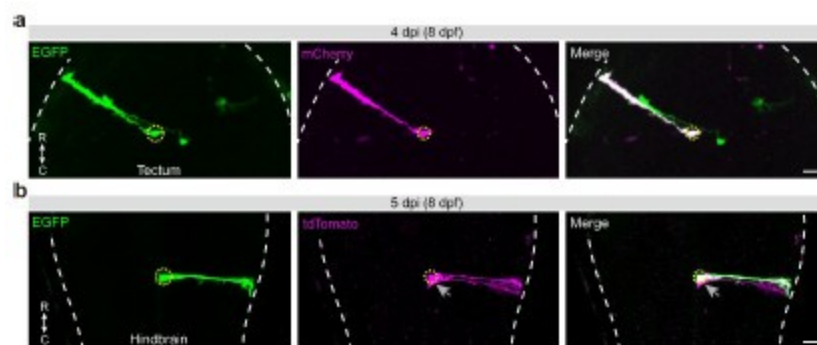
If TVA and rabies G are delivered to glial cells, followed by rabies virus injection, will it lead to the infection of other glial cells or neurons?

We have conducted experiments in which TVA and rabies G were specifically expressed in astroglia using the gfap promoter, followed by RVdG-mCherry[EnvA] injection. This resulted in initial infection of TVA-positive astroglia and occasional subsequent labeling of nearby TVA-negative astroglia (Author response image 4), suggesting astroglia-toastroglia transmission. Notably, no neuronal labeling was observed. This glial-to-glial spread is consistent with previous rabies tracing studies reporting similar phenomena involving the interaction of astrocytes with astrocytes and microglia (Clark et al., 2021). However, the underlying mechanism remains unclear, and we have discussed this in response to the first comment.

Author response image 4.

Viral tracing initiated from astroglia. (a) Confocal images of the tectum of a larva expressing EGFP and TVA using UGBT in randomly sparse astroglia (gfap⁺) and infected by SADdG-mCherry[EnvA] (magenta) injected into the anterior hindbrain. (b) Confocal images of the posterior hindbrain of a larva expressing EGFP and TVA using UGNT in randomly sparse astroglia (gfap⁺) and infected by CVSdG-tdTomato[EnvA] (magenta) injected into the anterior hindbrain. Dashed yellow circles, starter astroglia (EGFP⁺/mCherry⁺ or EGFP⁺/tdTomato⁺); gray arrows, transneuronally labeled astroglia (tdTomato⁺/EGFP⁺); dashed white lines, tectum

or hindbrain boundaries. C, caudal; R, rostral. Scale bars, 20 μ m.



Answers to any of these questions could greatly benefit the broader research community.

(2) The optimal virus tracing effect has to be achieved by raising the injected larvae at 36°C. Since the routine temperature of zebrafish culture is around 28°C, a more thorough characterization of the effect on the health of zebrafish should be conducted.

Yes, 36°C is required to achieve optimal labeling efficiency. Although this is above the standard zebrafish culture temperature (28°C), previous work (Satou et al., 2022) and our observations indicate that this transient elevation does not adversely affect larval health within the experimental time window.

In the previous study, Satou et al. reported no temperature-dependent effects on swimming behavior, social interaction, or odor discrimination in adult fish maintained at 28°C and 36°C. In larvae, both non-injected and virus-injected fish showed a decrease in survival at later time points (7 dpi), with slightly increased mortality observed at elevated temperatures.

In our study, we raised the same batch of non-virus-injected larvae at 28°C and 36°C, and found no mortality over a 10-day period. For CVS-N2c-injected larvae, electrode insertion caused injury, but survival rates remained around 80% at both temperatures (see Figure 3A). Moreover, we successfully maintained CVS-N2c-injected larvae at 36°C for over a month, indicating that elevated temperature does not adversely affect fish health. Notably, higher temperatures were associated with an accelerated developmental rate.

This point was briefly addressed in the previous version and has now been further elaborated in the revised Discussion section (page 8).

(3) Given the ability of time-lapse imaging of the infected larval zebrafish brain, the system can be taken advantage of to tackle important issues of rabies virus tracing tools.

a) Toxicity.

The toxicity of rabies viruses is an important issue that limits their application and affects the interpretation of traced circuits. For example, if a significant proportion of starter cells die before analysis, the traced presynaptic networks cannot be reliably assigned to a "defined" population of starter cells. In this manuscript, the authors did an excellent job of characterizing the effects of different rabies strains, G proteins derived from various strains, and levels of G protein expression on starter cell survival. However, an additional parameter that should be tested is the dose of rabies virus injection. The current method section states that all rabies virus preparations were diluted to 2×10^8 infection units per ml, and 2-5 nl of virus suspension was injected near the target cells. It would be interesting to know the impact of the dose/volume of virus injection on retrograde tracing efficiency and toxicity. Would higher titers of the virus lead to more efficient

labeling but stronger toxicities? What would be the optimal dose/volume to balance efficiency and toxicity? Addressing these questions would provide valuable insights and help optimize the use of rabies viruses for circuit tracing.

This is an important concern. Viral cytotoxicity is primarily driven by the level of viral transcription and replication, which inhibits host protein synthesis (Komarova et al., 2007). The RVdG-EnvA typically infects cells at a rate of one viral particle per cell (Zhang et al., 2024), suggesting that increasing viral concentration does not proportionally increase percell infection. Accordingly, viral titer and injection volume are unlikely to influence cytotoxicity at the single-cell level. In our experiments, injection volumes up to 20 nl (i.e., 4 to 10 times the standard injection volume) did not affect starter cell survival. However, higher titers or volumes may increase the number of initially infected starter cells, potentially leading to greater overall mortality in larval zebrafish.

Similarly, given that rabies virus typically infects cells at one particle per cell, increasing viral titer alone is unlikely to enhance tracing efficiency once the virus type is fixed. In contrast, the level of G protein expression significantly influences tracing efficiency (see Figure 2D). However, excessive G protein expression reduces the survival of starter cells (see Figure 3D). Therefore, careful control of G protein levels is essential to balance tracing efficiency and cytotoxicity.

Notably, regardless of whether infected cells undergo apoptosis or necrosis due to cytotoxicity, the resulting disruption of the plasma membrane severely impairs viral budding. As a result, the formation of intact, G protein-enveloped viral particles is prevented, limiting further infection of neighboring neurons.

The latest second-generation Δ GL RV vectors (Jin et al., 2024), which lack both the G and L (viral polymerase) genes, have been shown to markedly reduce cytotoxicity. These improved tracing strategies may be explored in future zebrafish studies to further optimize labeling efficiency and cell viability.

The issue of viral titer and volume has been addressed in the revised Discussion section (page 10).

b) Primary starters and secondary starters:

Given that the trans-expression of TVA and G is widespread, there is the possibility of coexistence of starter cells from the initial infection (primary starters) and starter cells generated by rabies virus spreading from the primary starters to presynaptic neurons expressing G. This means that the labeled input cells could be a mixed population connected with either the primary or secondary starter cells.

It would be immensely interesting if time-lapse imaging could be utilized to observe the appearance of such primary and secondary starter cells. Assuming there is a time difference between the initial appearance of these two populations, it may be possible to differentiate the input cells wired to these populations based on a similar temporal difference in their initial appearance. This approach could provide valuable insights into the dynamics of rabies virus spread and the connectivity of neural circuits.

The reviewers suggestion is valuable. Regarding the use of Purkinje cells (PCs) as starter cells, we consider the occurrence of secondary PCs to be extremely rare. Although previous evidence suggests that PCs can form synaptic connections with one another (Chang et al., 2020), our sparse labeling strategy—typically involving fewer than 10 labeled cells—significantly reduces the likelihood of viral transmission between PC starter cells. In addition, if secondary starter PCs were frequently generated, we would expect increased tracing

efficiency at 10 dpi compared to 6 dpi. However, our results show no significant difference (see Figure 2—figure supplement 1C and G).

Given the restricted expression of TVA and G in PCs, even if a limited number of secondary starters were generated, the labeled inputs would predominantly be granule cells (GCs), thereby preserving the cell-type identity of upstream inputs. While this raises a potential concern regarding an overestimation of the convergence index (CI). Notably, within the GC-PC circuit, individual GCs often project to multiple PCs. Consequently, a GC labeled via a secondary PC may also be a bona fide presynaptic partner of the primary starter population. This overlap could mitigate the overestimation of CI. Taken together, we believe that the CI values reported in this study provide a reasonable approximation of monosynaptic connectivity.

In scenarios where TVA and G are broadly expressed—for example, under the control of *vglut2a* promoter—secondary starter cells may arise frequently. In such cases, long-term time-lapse imaging in the zebrafish whole brain presents a promising strategy to distinguish primary and secondary starter cells, along with their respective input populations, based on the timing of their appearance. This approach potentially enables multi-step circuit tracing within individual animals. An alternative strategy is to use an EnvA-pseudotyped, G-competent rabies virus, which allows targeted initial infection while supporting multisynaptic propagation. When combined with temporally resolved imaging, this strategy could facilitate direct labeling of higher-order circuits and allow clear differentiation between multi-order inputs and the original starter population over time.

In conclusion, we find this suggestion compelling and will explore these strategies in future studies to optimize and broaden the application of rabies virus-based circuit tracing.

Reviewer #2 (Public Review):

The study by Chen, Deng et al. aims to develop an efficient viral transneuronal tracing method that allows efficient retrograde tracing in the larval zebrafish. The authors utilize pseudotyped-rabies virus that can be targeted to specific cell types using the EnvA-TvA systems. Pseudotyped rabies virus has been used extensively in rodent models and, in recent years, has begun to be developed for use in adult zebrafish. However, compared to rodents, the efficiency of the spread in adult zebrafish is very low (~one upstream neuron labeled per starter cell). Additionally, there is limited evidence of retrograde tracing with pseudotyped rabies in the larval stage, which is the stage when most functional neural imaging studies are done in the field. In this study, the authors systematically optimized several parameters of rabies tracing, including different rabies virus strains, glycoprotein types, temperatures, expression construct designs, and elimination of glial labeling. The optimal configurations developed by the authors are up to 5-10 fold higher than more typically used configurations.

The results are solid and support the conclusions. However, the methods should be described in more detail to allow other zebrafish researchers to apply this method in their own work.

Additionally, some findings are presented anecdotally, i.e., without quantification or sufficient detail to allow close examinations. Lastly, there is concern that the reagents created by the authors will not be easily accessible to the zebrafish community.

(1) The titer used in each experiment was not stated. In the methods section, it is stated that aliquots are stored at 2×10^8 . Is it diluted for injection? Are all of the experiments in the manuscripts with the same titer?

We injected all three viral vectors as undiluted stock aliquots. The titer for SAdGmCherry[EnvA], CVSdG-tdTomato[EnvA], and CVSdG-mCherry-2A-Cre[EnvA] was 2×10^8 , 2×10^8 , and 3×10^8 infectious units/mL, respectively. This has been clarified in the updated Methods section (page 12).

(2) The age for injection is quite broad (3-5 dpf in Fig 1 and 4-6 dpf in Fig 2). Given that viral spread efficiency is usually more robust in younger animals, describing the exact injection age for each experiment is critical.

We appreciate the reviewer's suggestions. For the initial experiments tracing randomly from neurons in Figure 1, the injection age was primarily 3–4 dpf, with a one-day difference. Due to the slower development of PCs, the injection age for experiments related to Figure 2,3, and 4, is mainly 5 dpf. To clarify the developmental stages at the time of injection for each experiment, we have newly added tables (see Figure 1,2—table supplement 2) listing the number of fish used at each injection age for all experimental groups shown in Figure 1 and 2.

(3) More details should be provided for the paired electrical stimulation-calcium imaging study. How many GC cells were tested? How many had corresponding PC cell responses? What is the response latency? For example, images of stimulated and recorded GCs and PCs should be shown.

Yes, these are important details for the paired electrical stimulation-calcium imaging study. We stimulated 33 GCs from 32 animals and detected calcium responses in putative postsynaptic PCs in 15 cases. Among these, we successfully ablated the single GC in 11 pairs and observed a weakened calcium response in PCs following ablation (see Figure 2M). The response latency was determined as the first calcium imaging frame where $\Delta F/F$ exceeded the baseline (pre-stimulus average) by 3 times the standard deviation. Imaging was performed at 5 Hz, and as shown in Figure 2L, the calculated average response latency was 152 ± 35 ms (mean $\pm$ SEM), indicating an immediate response with calcium intensity from the first post-stimulus imaging frame consistently exceeding the threshold.

We have added additional details to the Results (page 5), Discussion (page 9), and Methods (page 15) sections. A representative image showing both the stimulated GC and the recorded PC has been added to Figure 2 in the revised manuscript (see Figure 2K).

(4) It is unclear how connectivity between specific PC and GC is determined for single neuron connectivity. In other images (Figure 4C), there are usually multiple starter cells and many GCs. It was not shown that the image resolution can establish clear axon dendritic contacts between cell pairs.

In our experiments, sparse labeling typically results in 1–10 starter cells per fish. Regarding the case shown in Figure 4C (right column), only two PC starters were labeled, which simplifies the assignment of presynaptic inputs to individual PCs. Connectivity is determined based on clear axon-dendritic or axon-cell body apposition between GCs and PCs. We have accordingly added more details to the Methods (page 16) section regarding how we determined connectivity between specific PCs and GCs.

Reviewer #2 (Recommendations For The Authors):

To enable broader use of this technique, I would encourage the authors to submit their zebrafish lines, plasmids, and plasmid sequences to public repositories such as ZIRC and Addgene. Additionally, there is no mention of how viral vectors will be shared.

We have deposited the related zebrafish lines at CZRC (China Zebrafish Resource Center) and uploaded plasmid maps and sequences to Addgene. The viral vectors are available through BrainCase (Shenzhen, China). We have included the information in the revised manuscript.

Reviewer #3 (Public Review):

Summary:

The authors establish reagents and define experimental parameters useful for defining neurons retrograde to a neuron of interest.

Strengths:

A clever approach, careful optimization, novel reagents, and convincing data together lead to convincing conclusions.

Weaknesses:

In the current version of the manuscript, the tracing results could be better centered with respect to past work, certain methods could be presented more clearly, and other approaches worth considering.

Appraisal/Discussion:

Trans-neuronal tracing in the larval zebrafish preparation has lagged behind rodent models, limiting "circuit-cracking" experiments. Previous work has demonstrated that pseudotyped rabies virus-mediated tracing could work, but published data suggested that there was considerable room for optimization. The authors take a major step forward here, identifying a number of key parameters to achieve success and establishing new transgenic reagents that incorporate modern intersectional approaches. As a proof of concept, the manuscript concludes with a rough characterization of inputs to cerebellar Purkinje cells. The work will be of considerable interest to neuroscientists who use the zebrafish model.

Reviewer #3 (Recommendations For The Authors):

The main limitations of the work are as follows:

(1) The optimizations might differ for different neurons. Purkinje cells are noteworthy because they develop considerably during the time window detailed here, almost doubling in number between 7-14dpf. Presumably, connectivity follows. This sort of neurogenesis is much less common elsewhere. It would be useful to show similar results in, say, tectal neurons, which would have spatially-restricted retinal ganglion cells labelled.

We acknowledge that Purkinje cells (PCs) undergo significant development between 7–14 dpf, which may influence synaptic connectivity and result in differences in tracing efficiency. However, all experimental conditions were standardized across groups, and the selection of starter PCs was unbiased, typically focusing on PCs in the lateral region of the CCe (corpus cerebelli) subregion, ensuring that the relative comparisons remain valid.

We agree that testing other neuronal populations would be valuable, as tracing efficiency is influenced by multiple factors, such as the number of endogenous inputs, synaptic maturation, and developmentally regulated synaptic strength. Tectal neurons, which receive spatially restricted retinal ganglion cell inputs, would be a suitable choice for further investigation. However, due to the various tectal cell types and the opacity of the eyeball, such studies present additional technical challenges and are beyond the scope of this paper.

(2) The virus is delivered by means of microinjection near the cell. This is invasive and challenging for labs that don't routinely perform electrophysiology. It would be useful to know if coarser methods of viral delivery (e.g. intraventricular injection) would be successful.

Our protocol does not require the level of precision needed for electrophysiology. The procedure can be performed using a standard high-magnification upright (135× magnification, Nikon SMZ18) or inverted fluorescence microscope (200× magnification, Olympus IX51). The virus suspension was loaded into a glass micropipette with a ~10 μm tip diameter and directly microinjected into the target region using a micromanipulator. The procedure was comparable to embryonic microinjection in terms of precision and operational control. Notably, direct contact with the target cells is not necessary, as the injected virus solution can diffuse and effectively infect nearby cells.

We had attempted intraventricular injection as an alternative, but it failed to produce robust labeling, reinforcing the necessity for direct tissue injection.

We have now included additional methodological details in the Methods section (page 13).

(3) Because of the combination of transgenic lines, plasmid injection, and viral type, it is often confusing to follow exactly what is being done for a particular experiment. It would be useful to specify the transgenic background used for each experiment using standard nomenclature e.g. "Plasmids were injected into Tg(elavl3:GAL4) fish." This is particularly important for the experiments in Figure 4: it isn't clear what the background used for the sparse labels was.

Thank the reviewer for bringing this issue to our attention. In order to improve clarity, we have revised the figure legends to explicitly state the transgenic background, injected plasmids, and viral type used in each experiment, particularly for Figure 4.

(4) Plasmids should be deposited with Addgene along with maps specifying the particular "codon-optimized Tetoff" per 388.

We confirm that all plasmids, including those containing codon-optimized Tetoff constructs, have been uploaded to Addgene along with detailed maps.

(5) It would be useful to know if there were more apoptotic cells after transfection -- an acridine orange or comparable assay is recommended, rather than loss of fluorescence.

We appreciate the reviewer's suggestion to assess apoptosis using acridine orange staining or comparable assays. We agree that such methods can provide more direct detection of apoptotic events. However, we believe that the difference in cytotoxicity is already evident in our current data: SAD-infected cells exhibit greater loss than CVS-infected cells (see Figure 3D). This is consistent with previous observations in mice, where greater toxicity of SAD compared to CVS was demonstrated using propidium iodide (PI) staining in cultured cells (Reardon et al., 2016).

(6) Line 219-228 Hibis lab has described the subtypes of granule cells in detail already; the work should discuss the tracings with respect to previous characterizations instead of limiting that work to a citation.

Thanks for the reminding of this point. We have expanded the Results section (page 6) to discuss the subtypes of GCs and PCs in relation to previously reported characterizations.

(7) "Activities" is often used when "activity" is correct. The use of English in the manuscript is, by and large, excellent, but its worth running the text through software like Grammarly to catch the occasional error.

We have carefully edited the manuscript using professional language editing tools to correct any grammatical issues.

(8) The experiments in 2J-2L would be more convincing if they were performed on inferior olive inputs as well -- especially given the small size of the granule cells.

We acknowledge the reviewers observation that granule cells (GCs) are relatively small, which may underline the finding that, out of 33 stimulated GCs, only 15 were capable of eliciting calcium responses in putative postsynaptic PCs. However, in all 11 pairs where a single GC was successfully ablated, we observed a weakened calcium response in PCs after the ablation (see Figure 2M), suggesting our tracing approach specifically identifies synaptically coupled neurons. We have clarified this point in the revised manuscript (page 5).

We agree that verifying the IO inputs to PCs would strengthen the validity of our findings. However, in our experiments, the probability of tracing upstream IO cells was relatively low. This may be due to the developmental immaturity of the synapse and the fact that each PC typically receives input from a single IO cell. Additionally, the deep and distant anatomical location of the IO presents technical challenges for paired electrical stimulationcalcium imaging study. To address these limitations, we are currently exploring the integration of viral tracing and optogenetics to further investigate IO-PC connectivity in future studies.

(9) It would be useful if the manuscript discussed the efficacy of trans-synaptic labelling. What fraction of granule cell / olivary inputs to a particular Purkinje cell do the authors think their method captures?

This is an important point for assessing the efficacy of our trans-synaptic labeling. Ideally, electron microscopy (EM) data would provide the most precise evaluation. In the absence of EM data, we estimated the number of GCs, IOs and PCs using light microscopy-based cell counting.

At approximately 7 dpf, we manually counted 327 ± 14 PCs and 2318 ± 70 GCs in the Tg(2 $\times$ en.cpce-E1B;tdTomato-CAAX) and Tg(cbln12:GAL4FF);Tg(5 $\times$ UAS:EGFP) zebrafish cerebellum, across all subregions (Va, CCe, EG, and LCa). Given the developmental increase in the number of GCs and the fact that some GCs that have exclusively ipsilateral projections, and that a single PC would not receive input from all parallel fibers, we estimate that by 10–14 dpf, a single PC receives approximately 1000–2000 GC inputs. Under optimal tracing conditions, we observed an average of 20 labeled GC inputs per PC, yielding a capture fraction of ~1–2%. Although this represents only a subset of total inputs, it is consistent with mammalian studies (Wall et al., 2010; Callaway et al., 2015), suggesting inherent limitations of this viral labeling approach.

For IO inputs, we counted 325 ± 26 inferior olivary neurons in Tg(elavl3:H2B-GCaMP6s) fish. A single PC likely receives input from one IO neuron, though an IO neuron may innervate multiple PCs. Accordingly, the observed capture rate for IO inputs was lower (7 out of 248 starters).

Further optimization is required to enhance the tracing efficiency. We have now incorporated a Discussion on this point in the revised manuscript (page 8).

<https://doi.org/10.7554/eLife.100880.2.sa0>